Napari

Introduction

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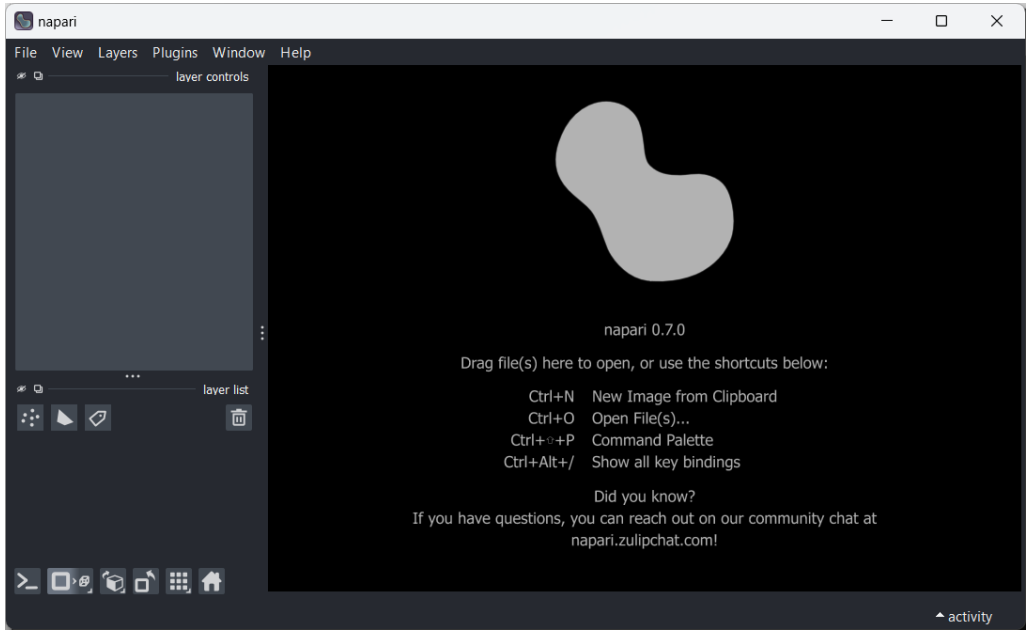
00

What is Napari ?

Napari

- Fast, interactive, N-dim image viewer written in **Python.**
- Layer-Based Architecture (Similar to Photoshop)
- Support natively **NumPy** arrays, **Dask** arrays, or **Zarr**
- Extensible **Plugin** Mechanism
- Interactive **GUI** & **Terminal**
- **GPU**-accelerated



01

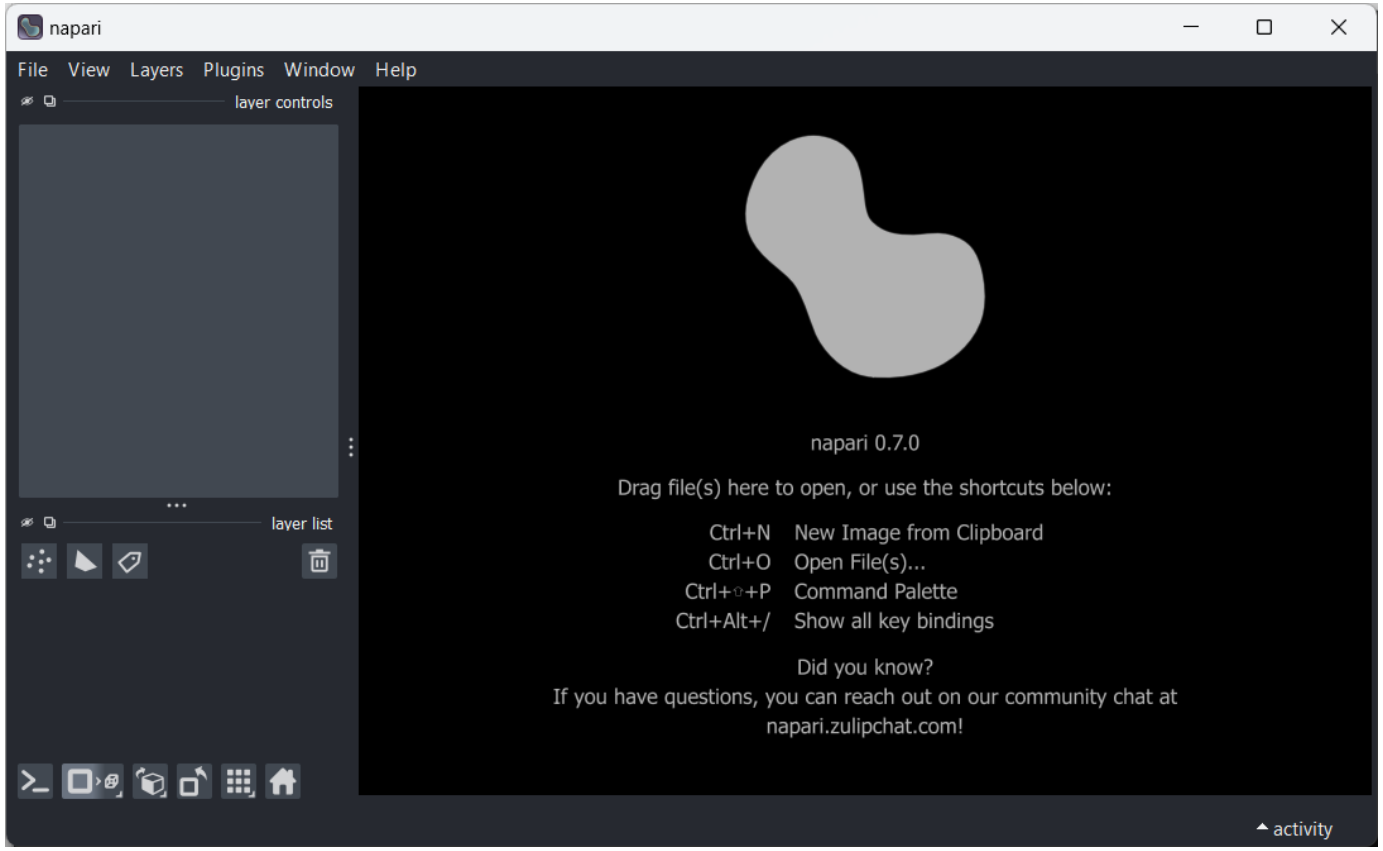
Installation & Run

Pre-requirement

- Conda installed (e.g via. Miniforge)
- Venv
- pixi

Conda

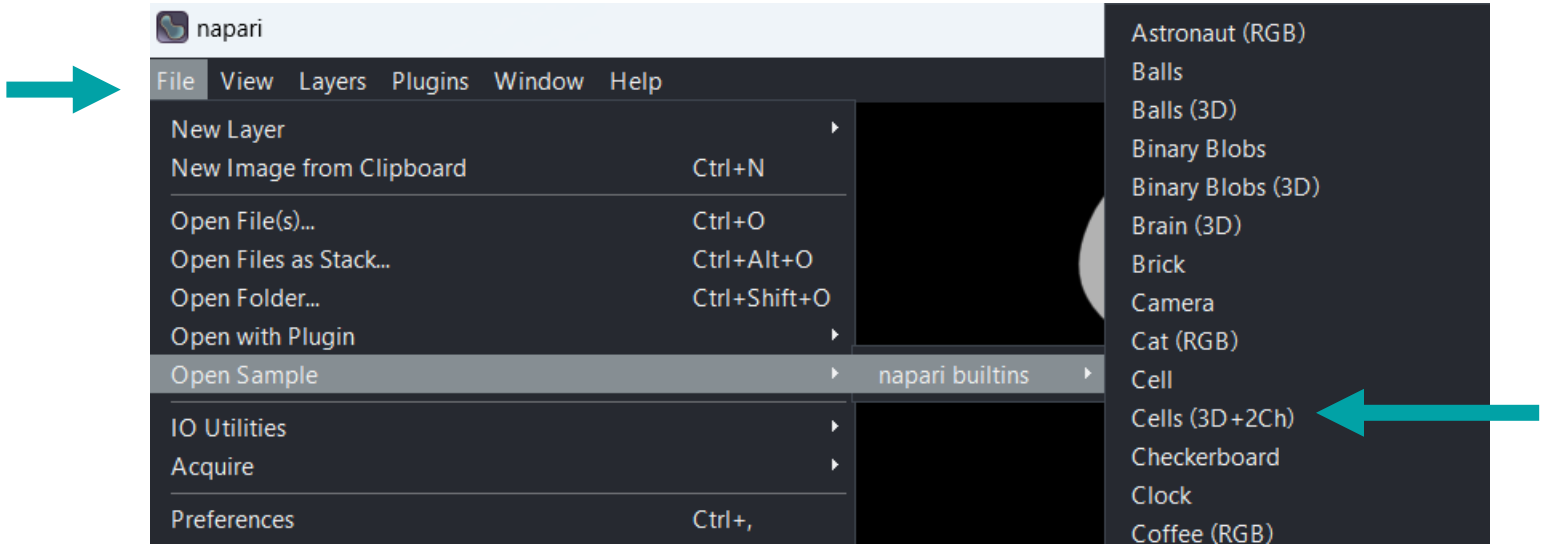
```
conda create -n napari-env python=3.12 napari pyqt
conda activate napari-env
napari
```



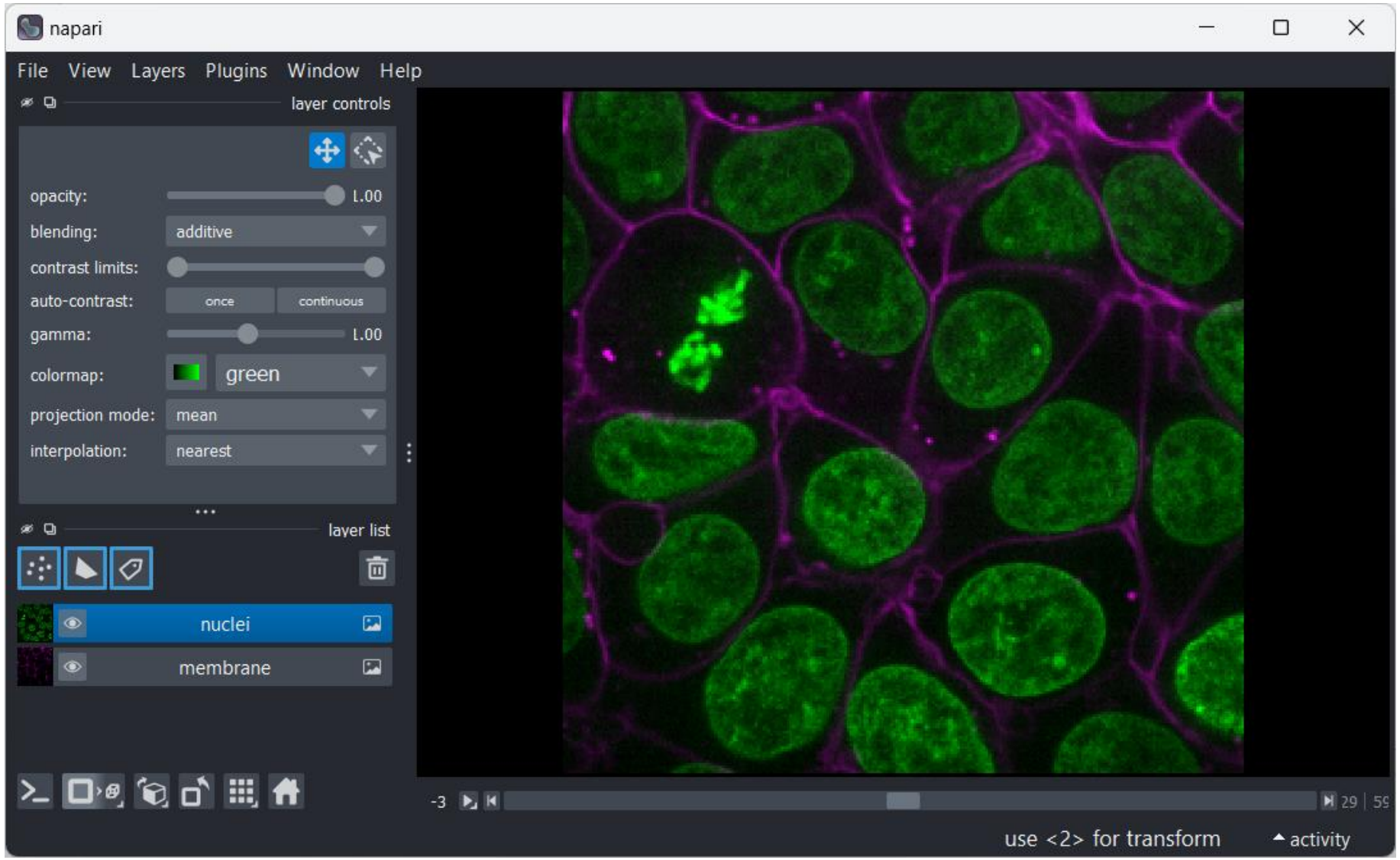
Napari

- Open demo Image:

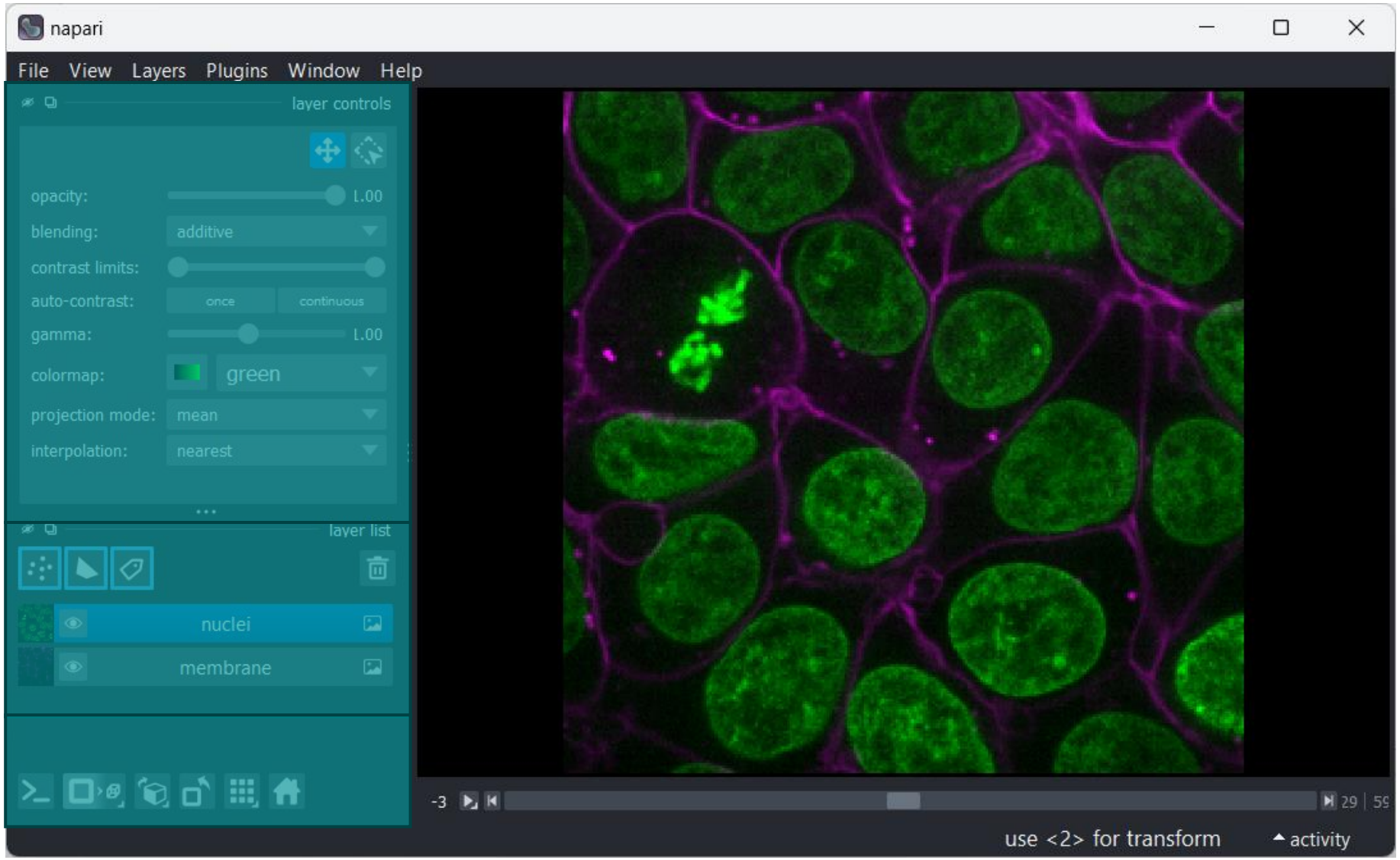
File > Open Sample > napari built-in > Cells (3D+2Ch)



Napari


napari
File
View
Layers
Plugins
Window
Help
layer controls
opacity:
1.00
blending:
additive
contrast limits:
auto-contrast:
once
continuous
gamma:
1.00
colormap:
green
projection mode:
mean
interpolation:
nearest
layer list
nuclei
membrane
-3
29
use <2> for transform
activity

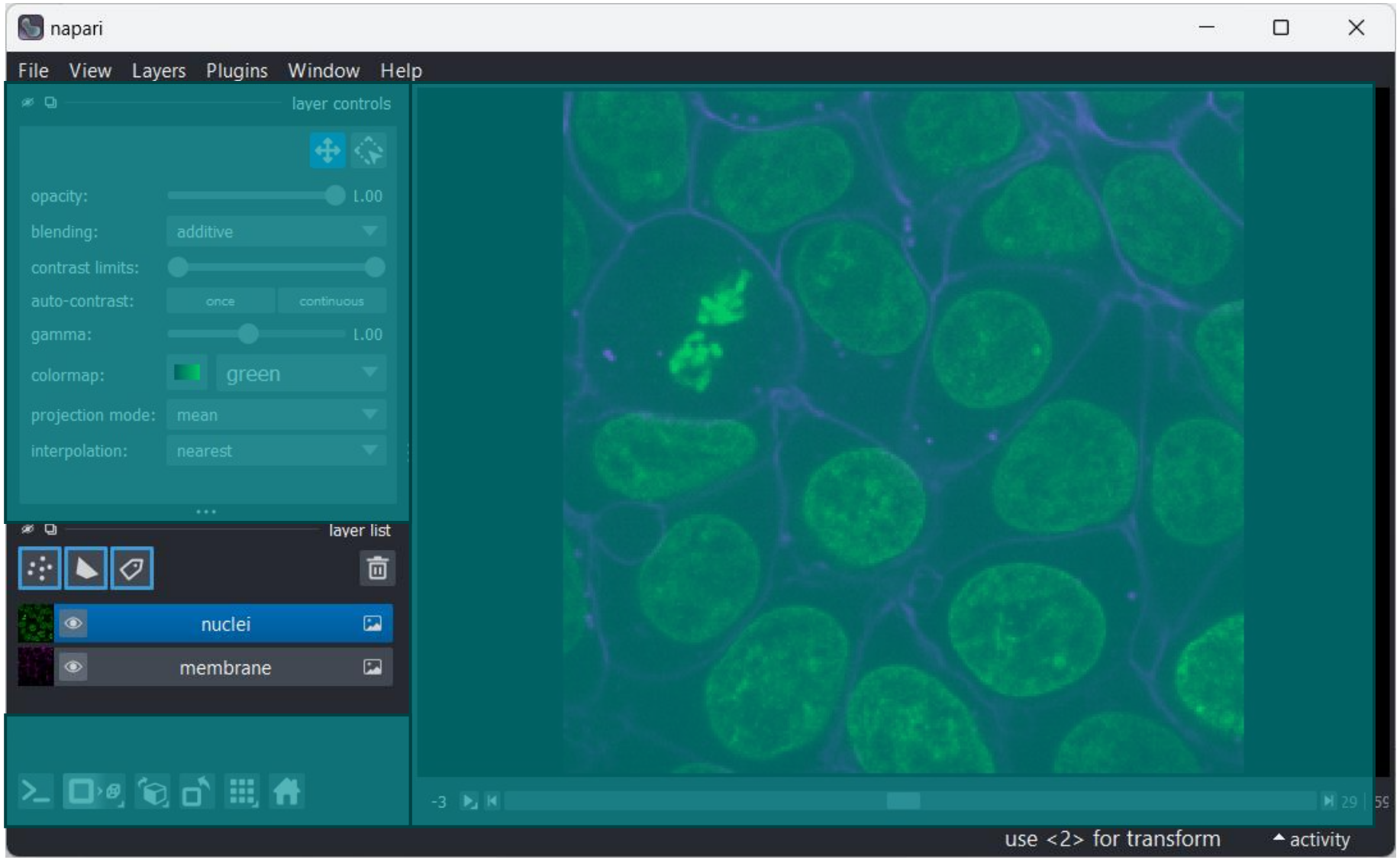
Napari



Viewer

VIB

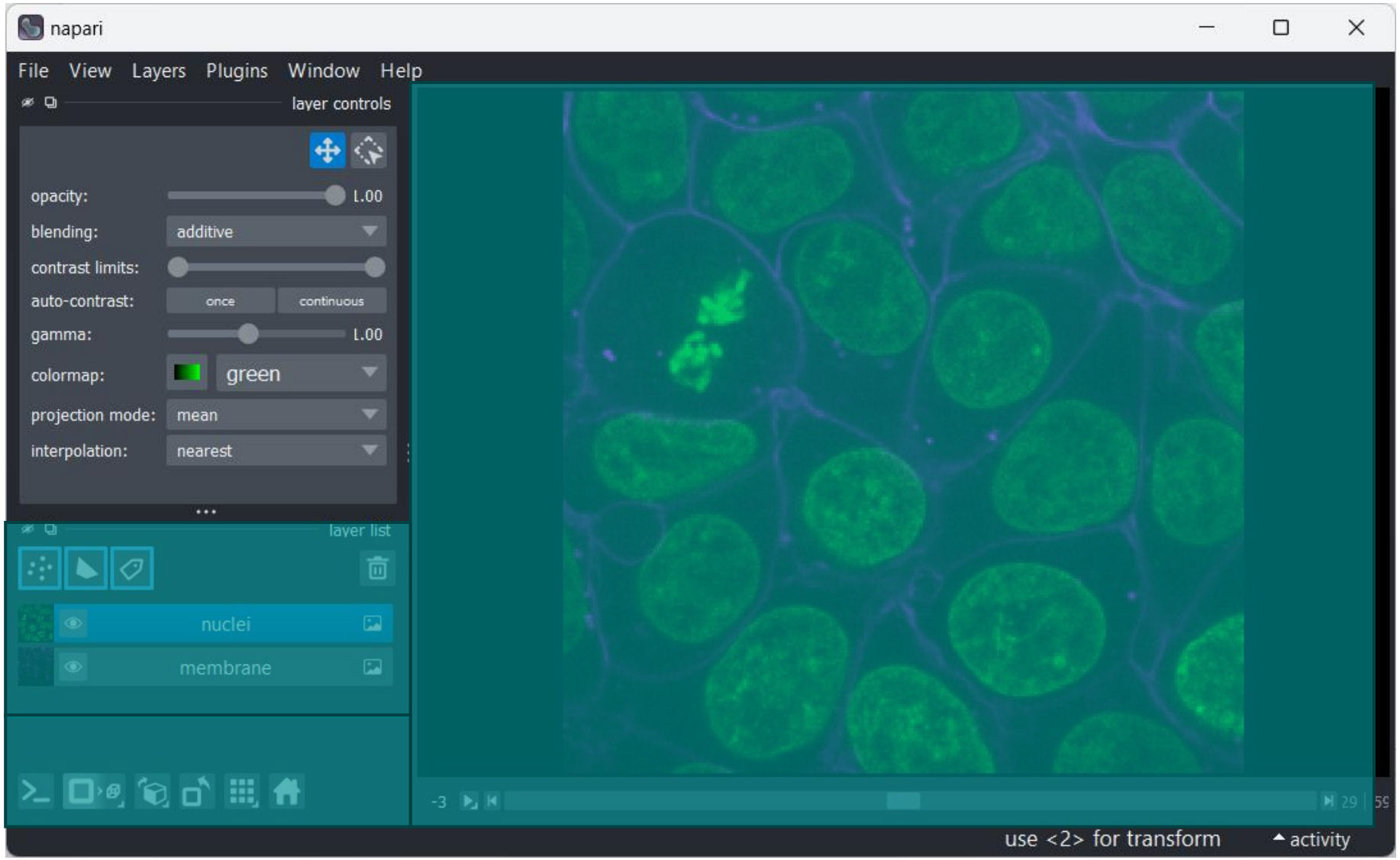
Napari



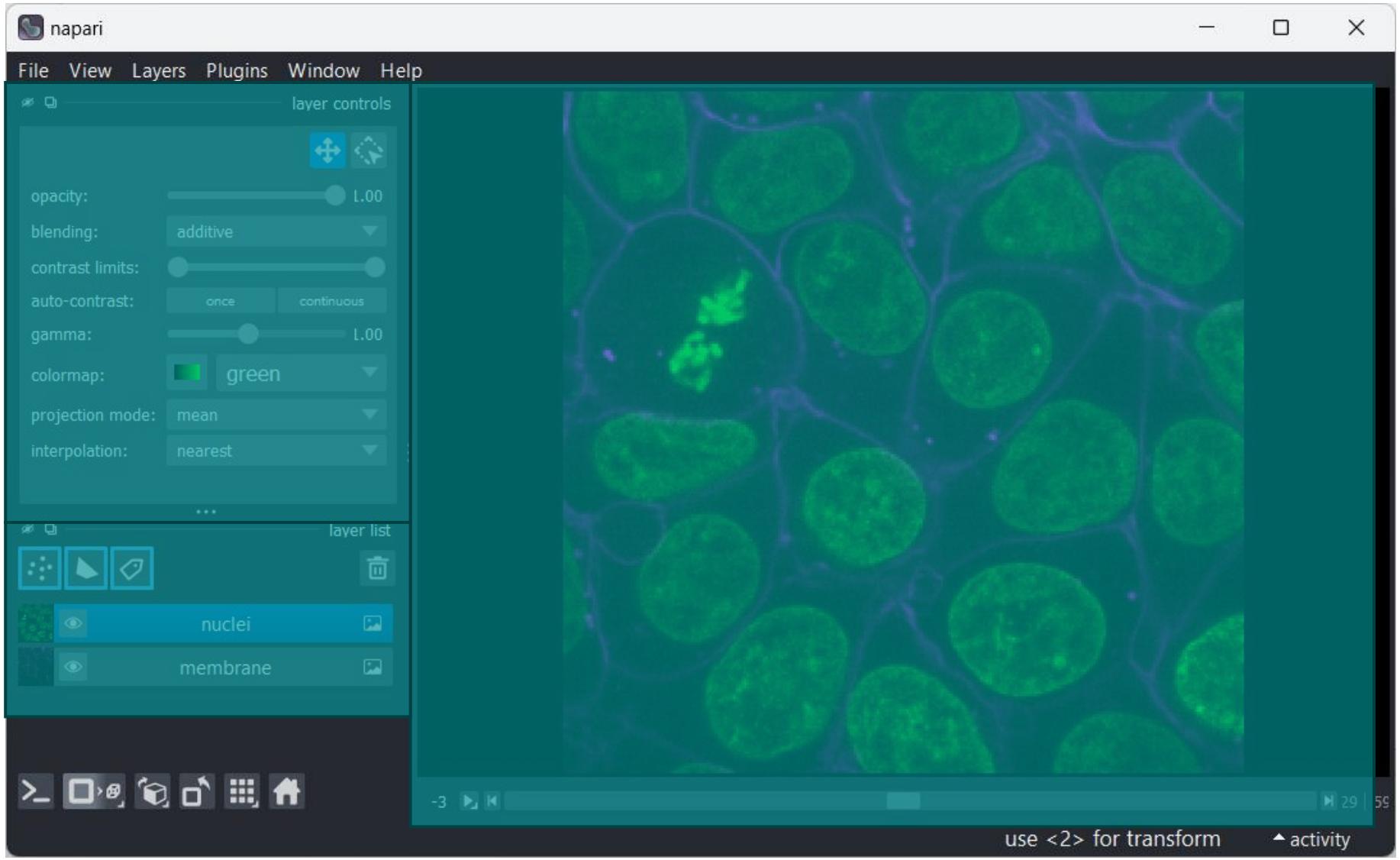
Layer List

Napari

Layer Control



Napari



Tool bar

Napari

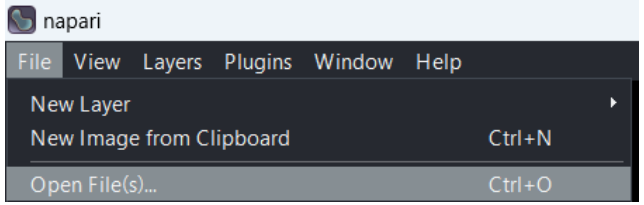
- Open a terminal to interact with the viewer using Python code
- Switch view
- 2D/3DChange the dimension for rendering
- Flip the view
- Show all layer as Grid
- Reset the view

VIB

Napari : Image

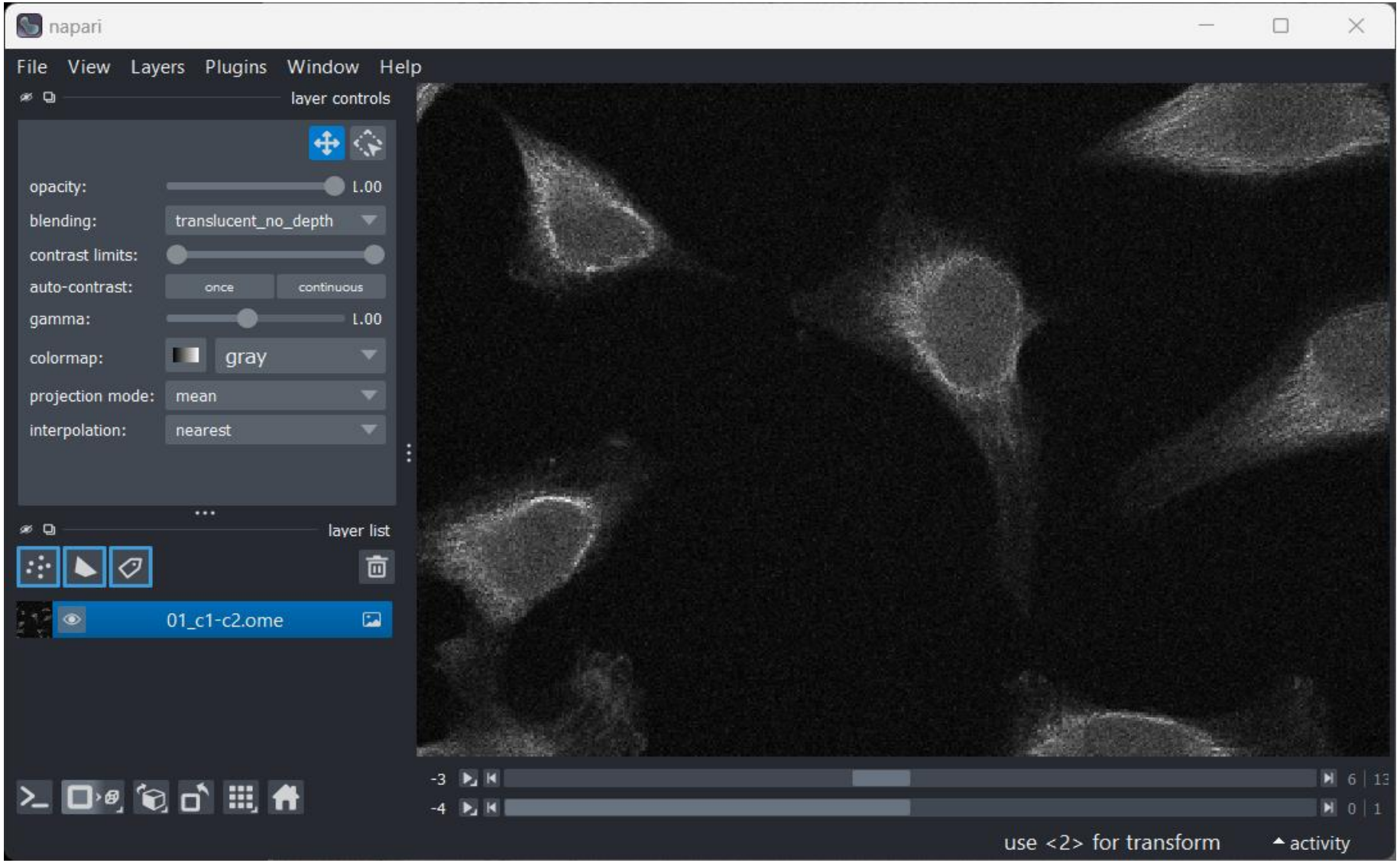
`data/01_hela_prolongdiamond_dapi_tubulin.ome.tiff`

`File > Open File(s)…`



Select the tiff file
`data/01_hela_prolon...`

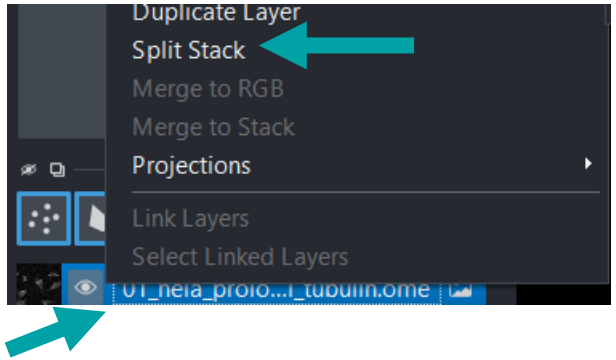
Unknown dimension,
Double click to rename
Them
(1st is Z, 2nd is C)



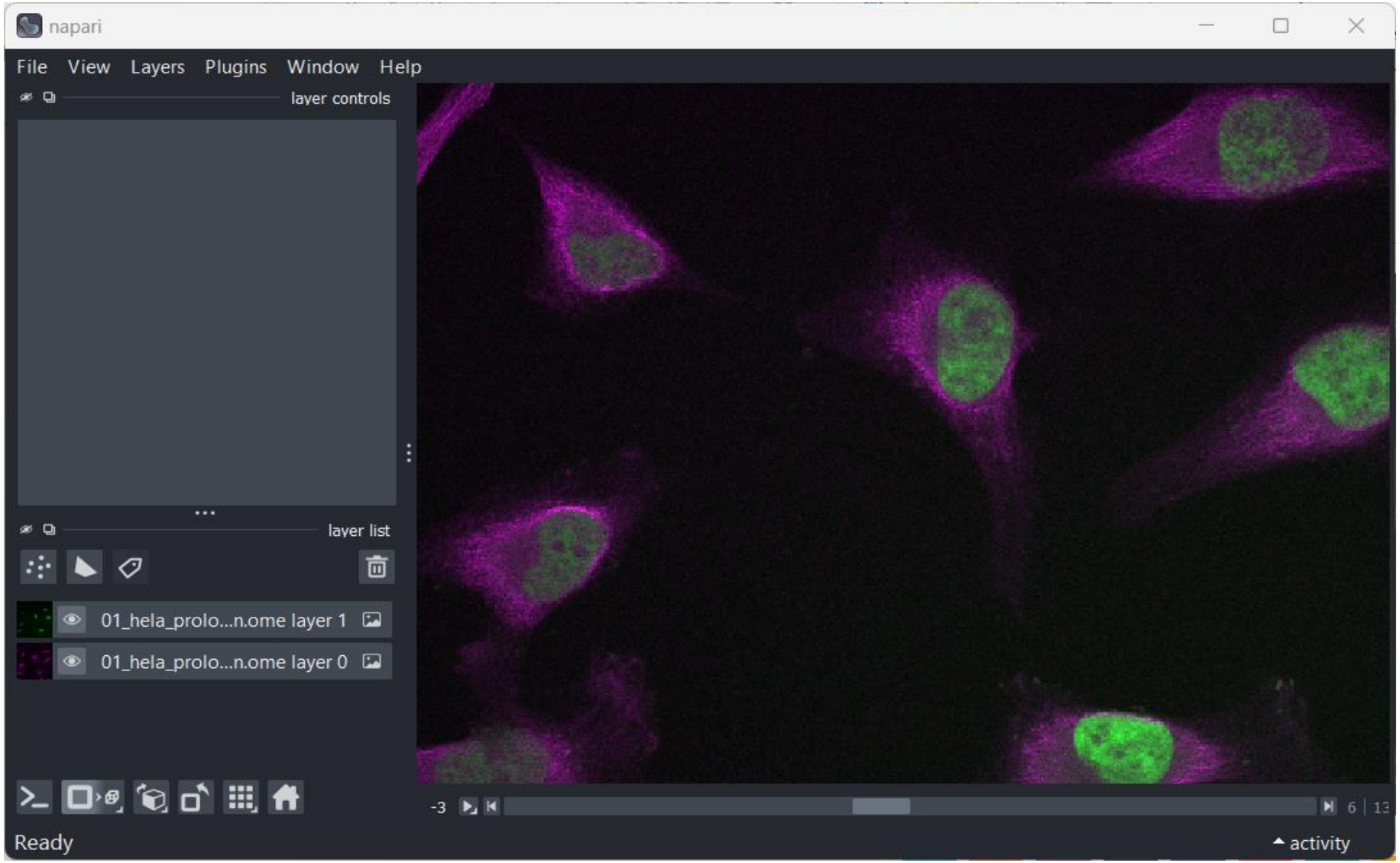
Napari : Image

data/01_hela_prolongdiamond_dapi_tubulin.ome.tiff

Right Click on the layer
> Split Stack



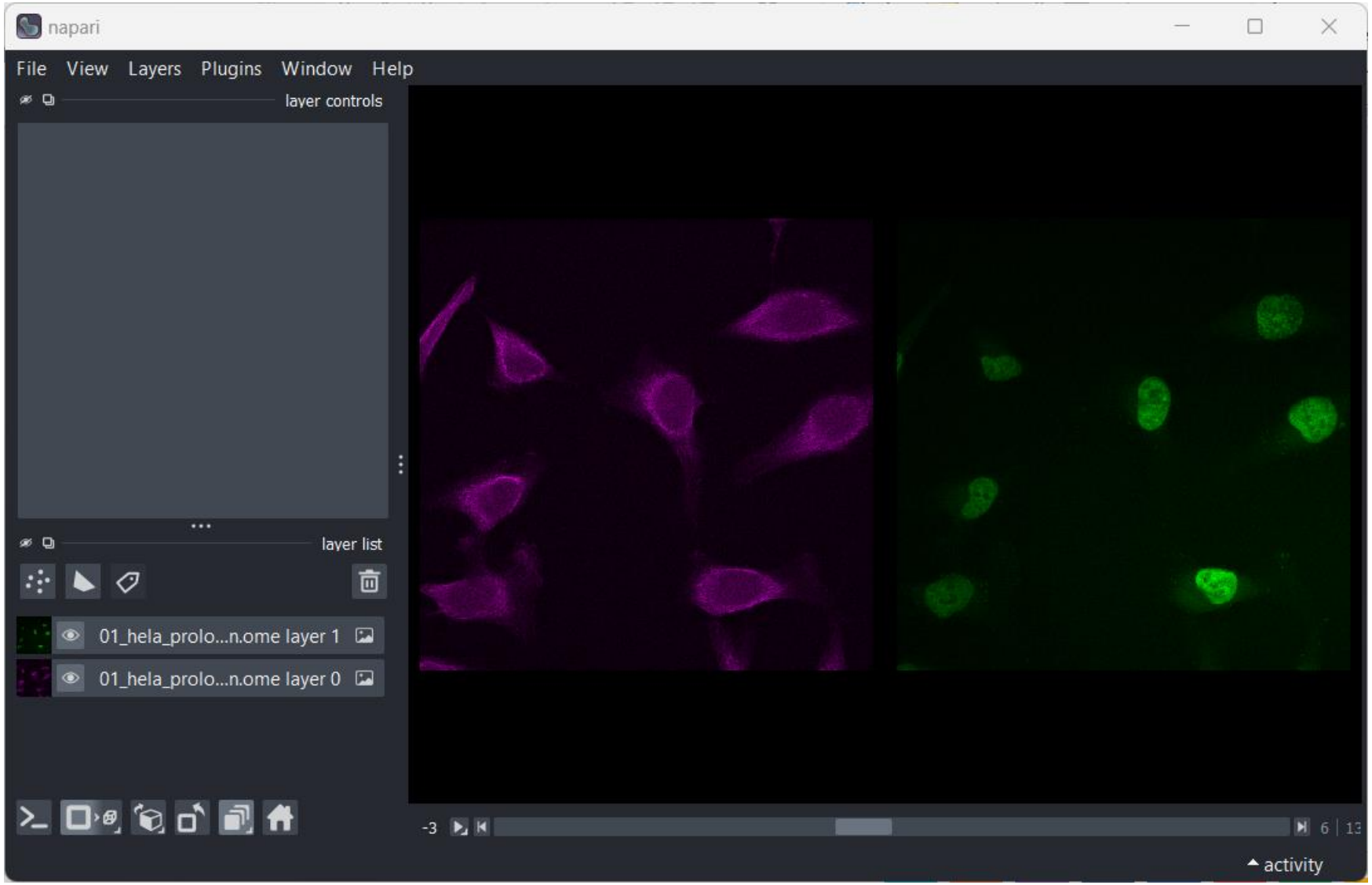
Modify the layer name by double clicking on the layer



Napari : Image

data/01_hela_prolongdiamond_dapi_tubulin.ome.tiff

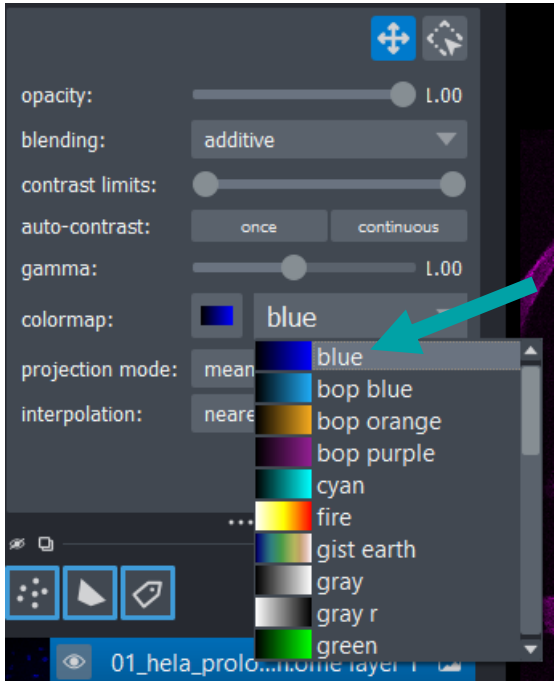
Split View

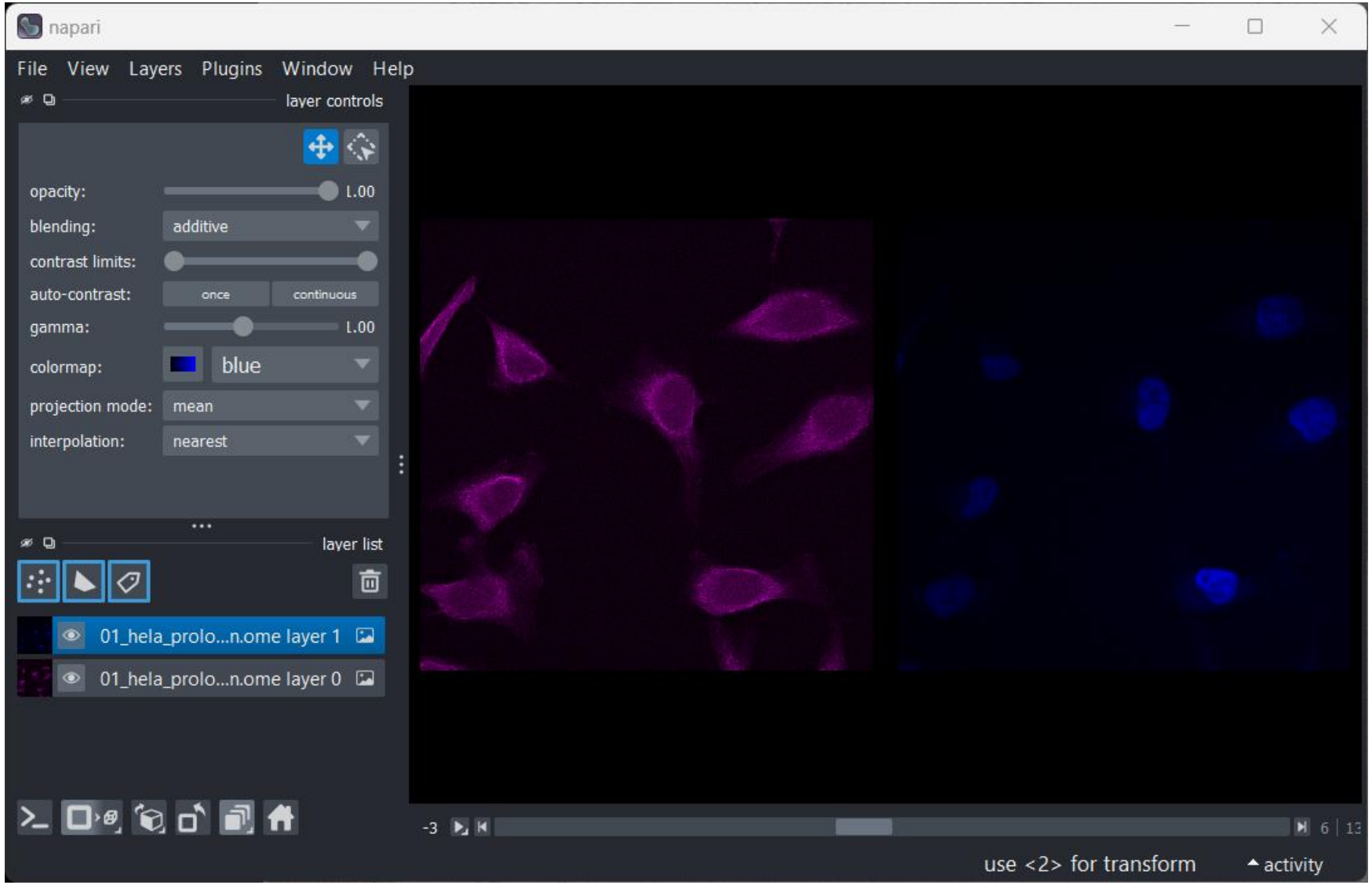


Napari : Image

data/01_hela_prolongdiamond_dapi_tubulin.ome.tiff

Change the rendering

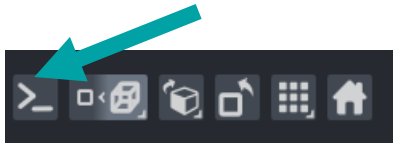




data/01_hela_prolongdiamond_dapi_tubulin.ome.tiff

Napari : Image Scaling

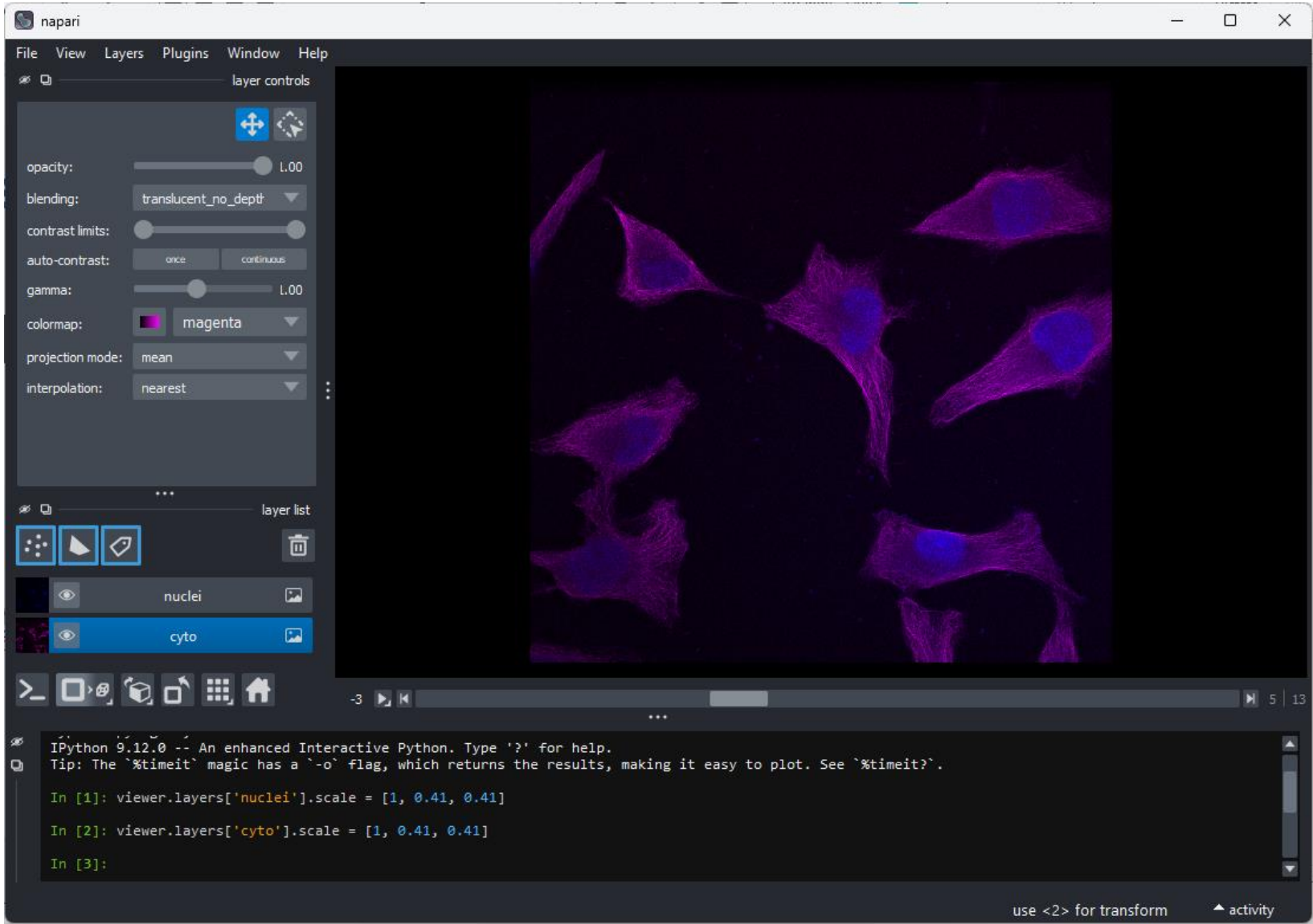
Console



The pixel size here is:
X/Y : 0,41 µm
Z : 1 µm

Set the scaling with the console :

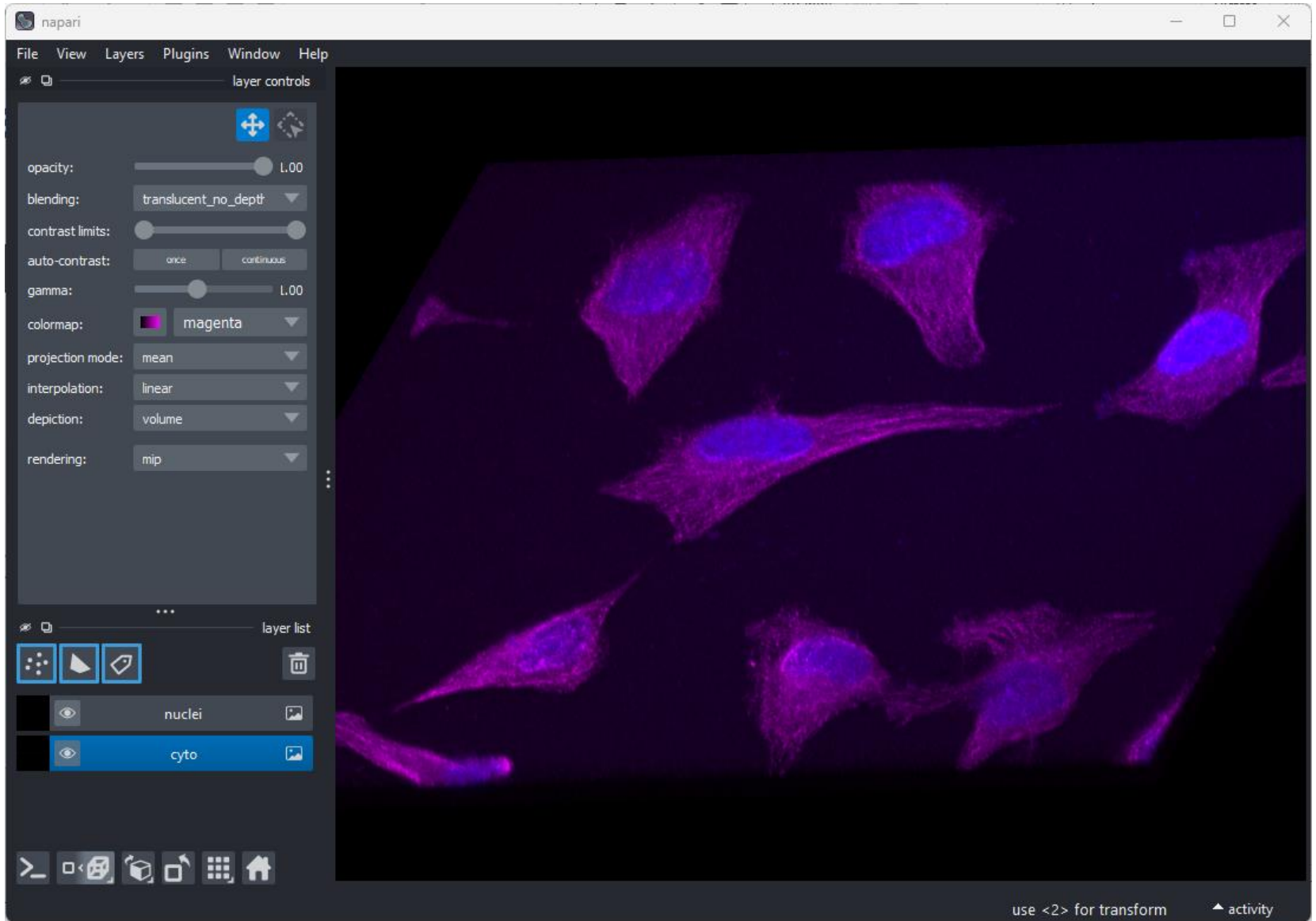
```
viewer.layers['nuclei'].scale = [1, 0.41, 0.41]
viewer.layers['cyto'].scale = [1, 0.41, 0.41]
```



Napari : Image

data/01_hela_prolongdiamond_dapi_tubulin.ome.tiff

3d Viewer



Napari : Image

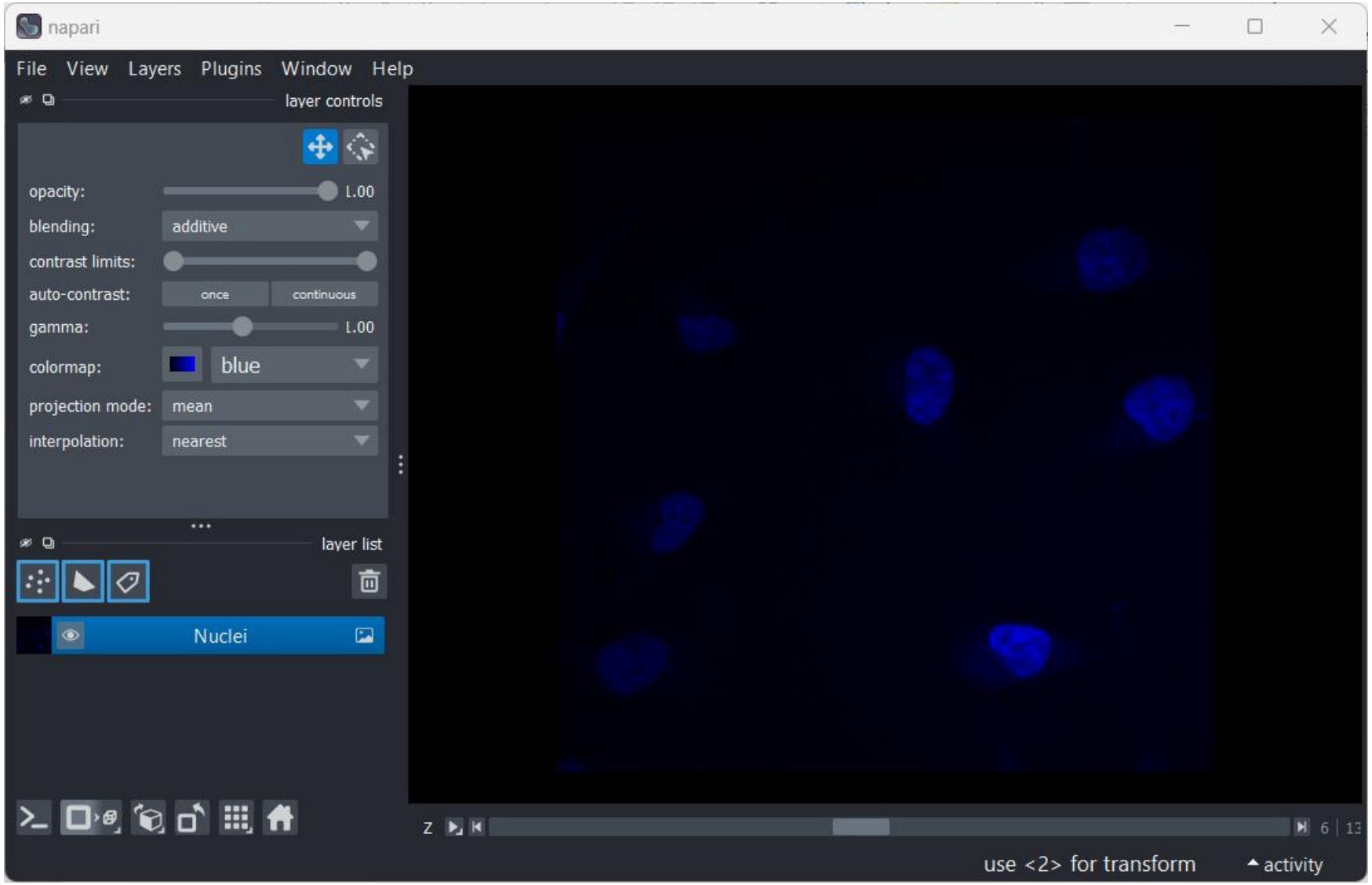
Exercice : Play a bit with the Image Viewer to test the different possibility (gamma, contrast, renaming layers etc…)

Napari : Images

Delete the cytoplasmic Channel

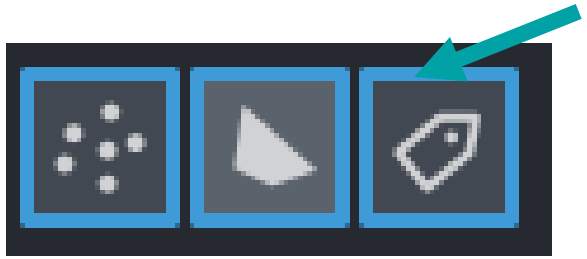
Set the colormap to blue

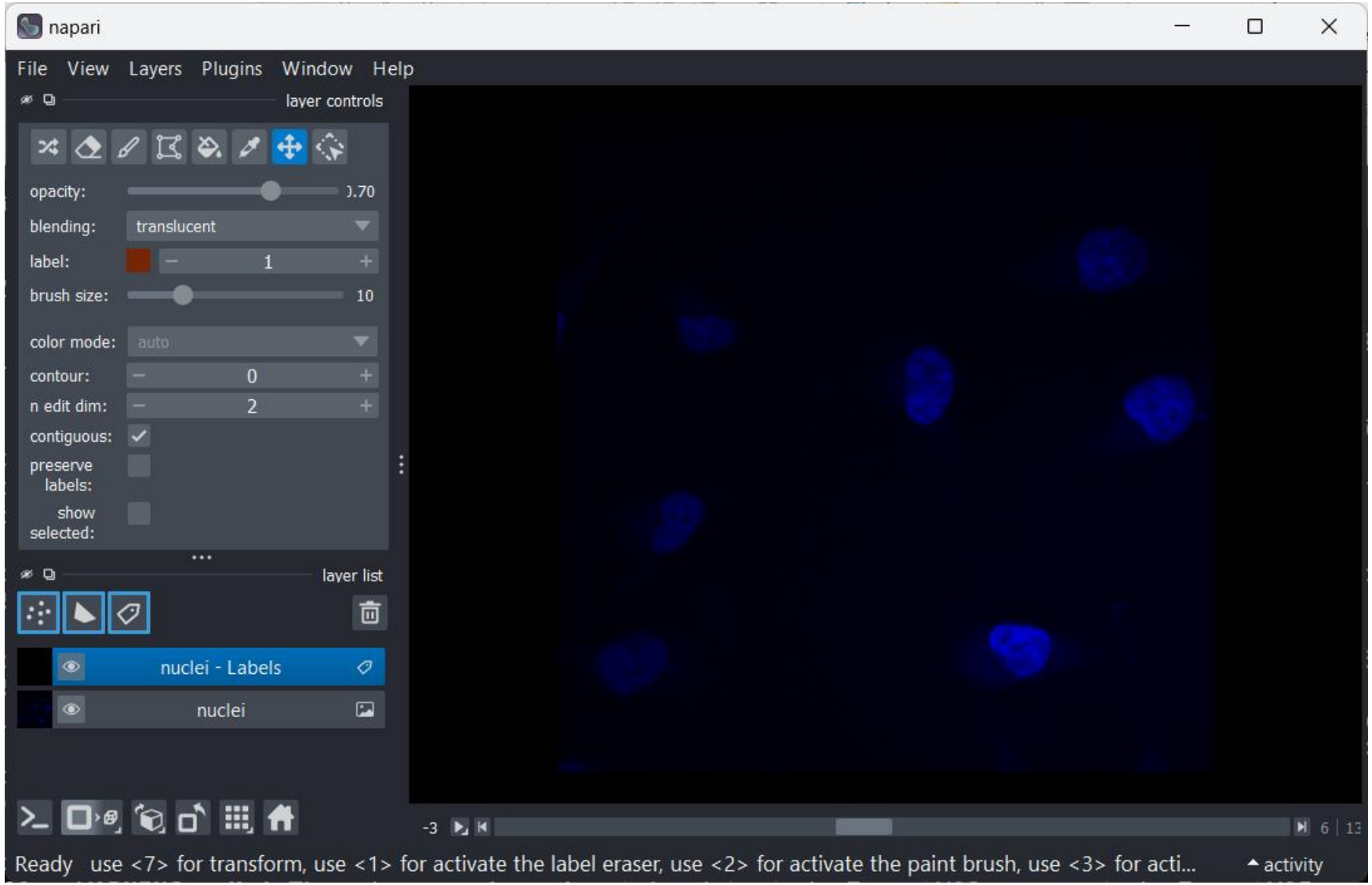
Rename the layer to nuclei



Napari : labels

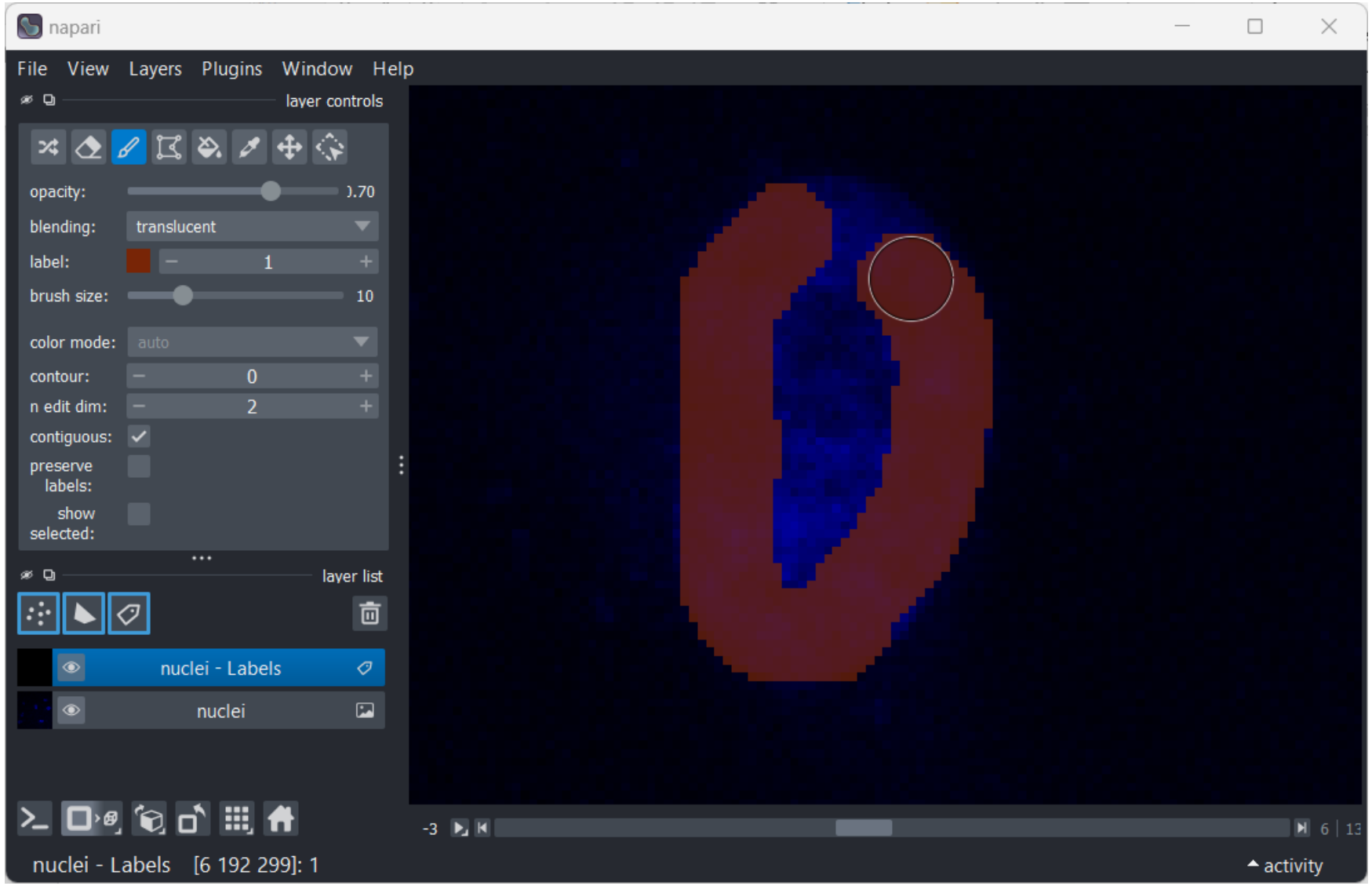
Create a label layer





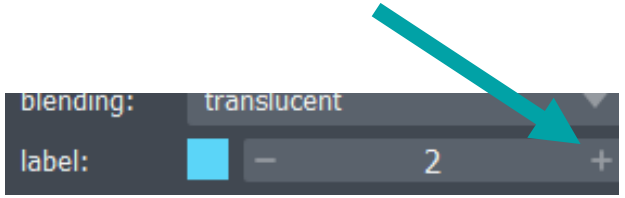
Napari : labels

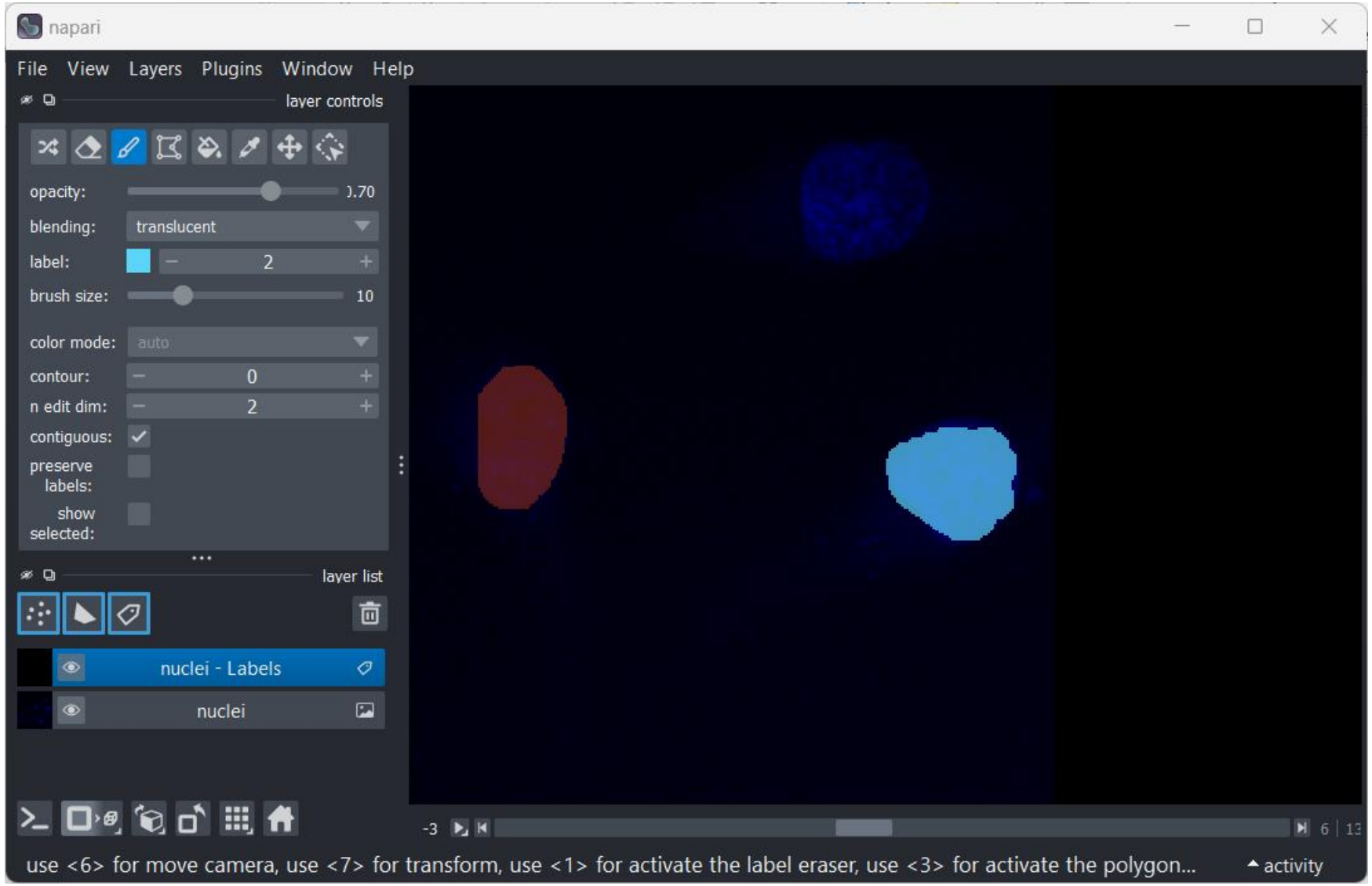
Select Pencil tool
and start to draw



Napari : labels

Select label 2 and draw the 2nd nuclei

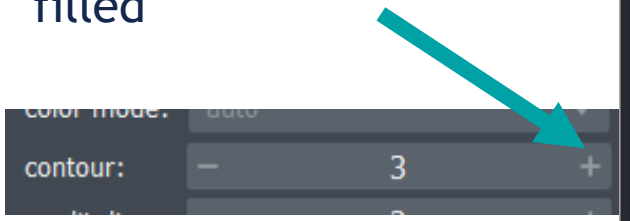


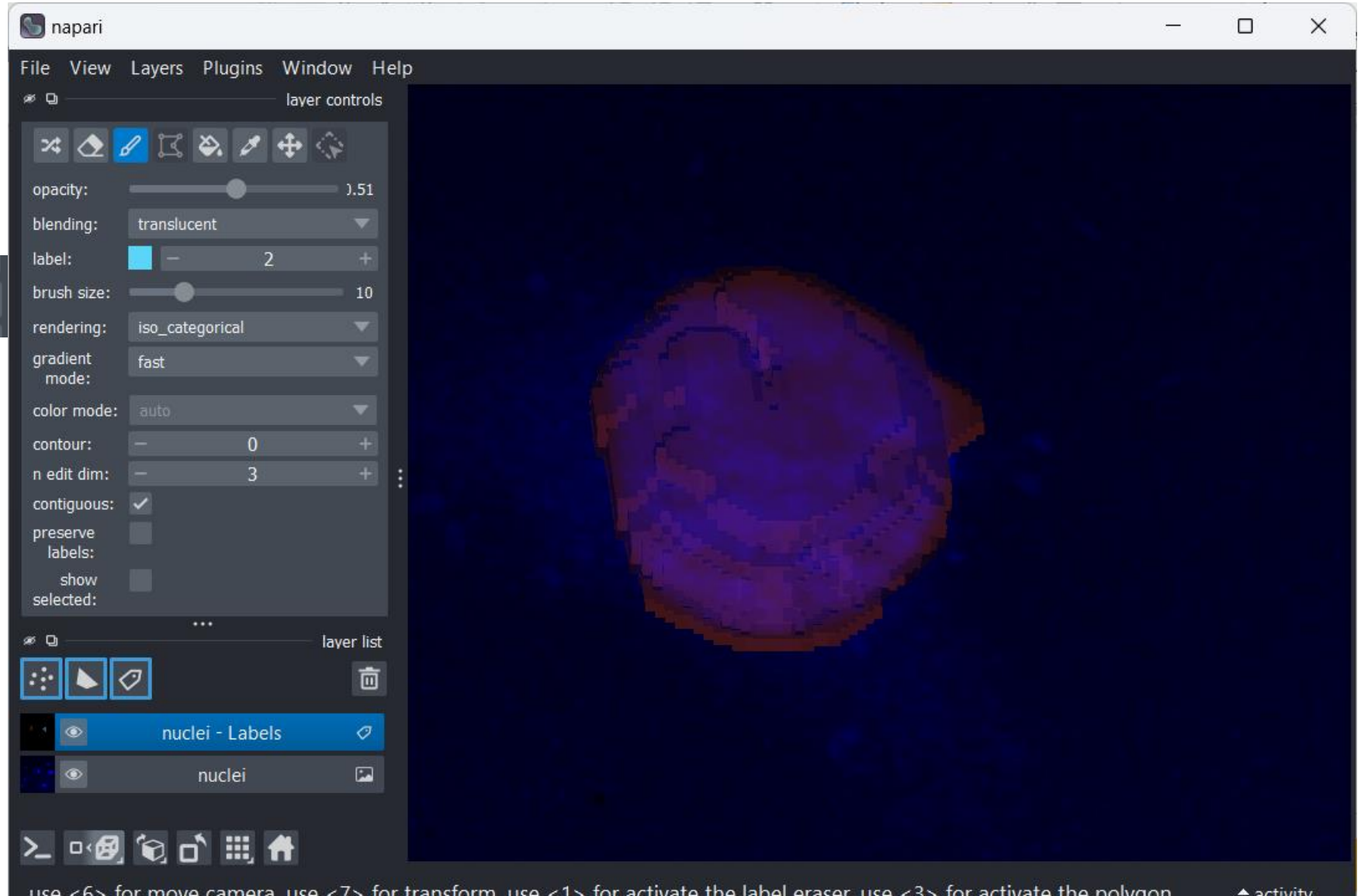


We just segmented 2 nuclei in 2D !

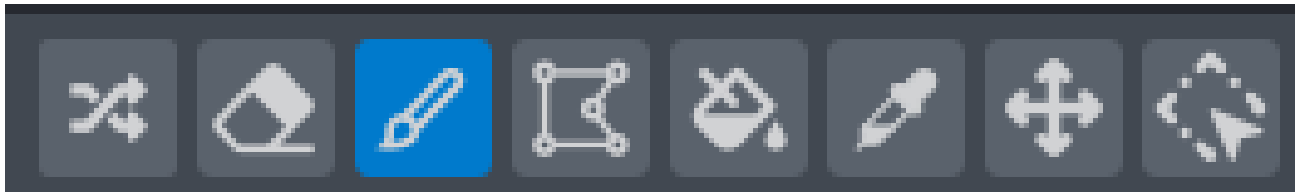
Napari : labels

Set a contour value to display contour instead of filled





Napari : label tools



Randomize the label colors

Erase

Fill. Select a label 0, then use it to erase a label by clicking on it

Pick

Moves the *camera*

Move the layer in *space*

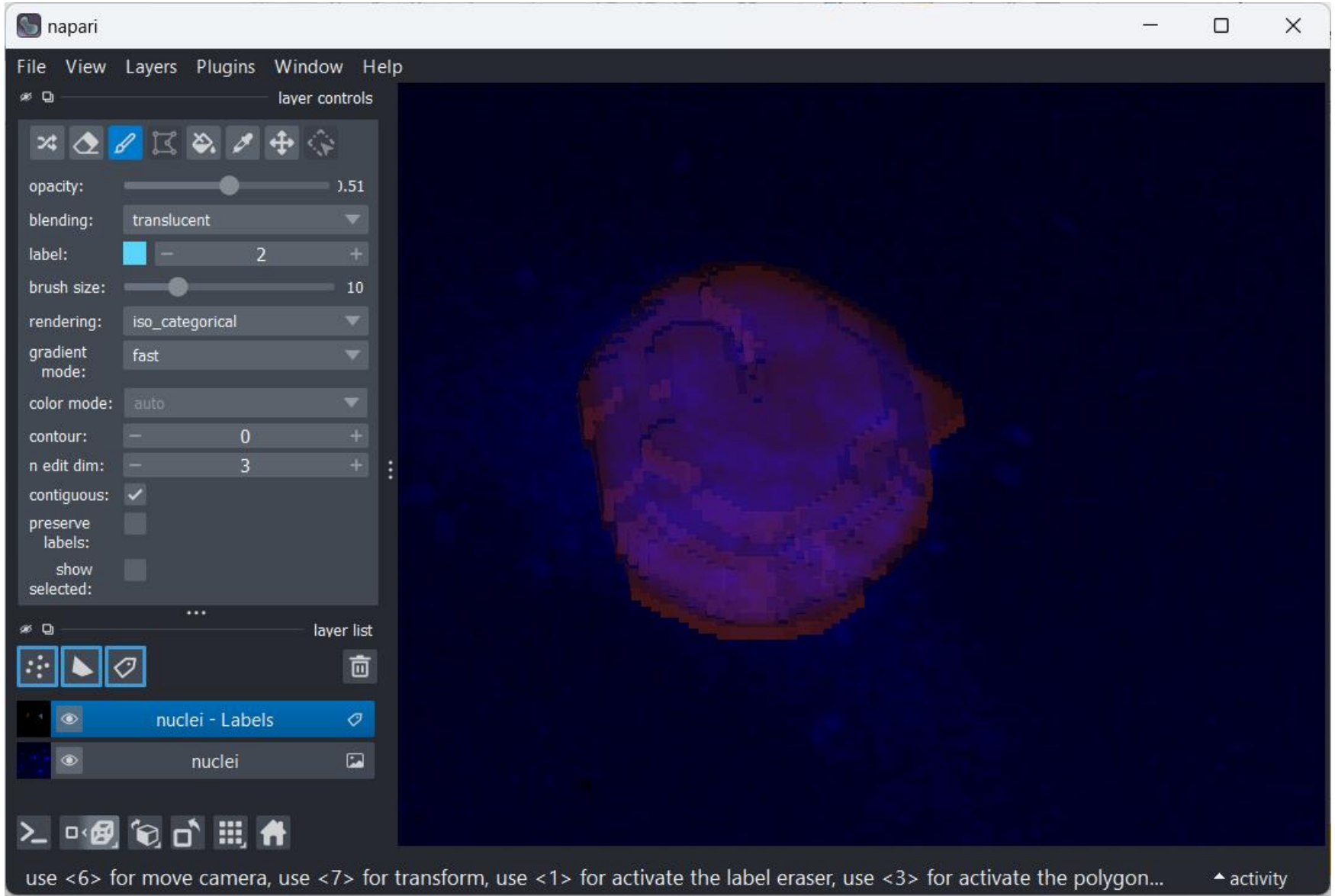
The brush size in the interface is limited to 40px, you can increase it via the console

```
viewer.layers['labels'].brush_size = 100
```

Napari : labels

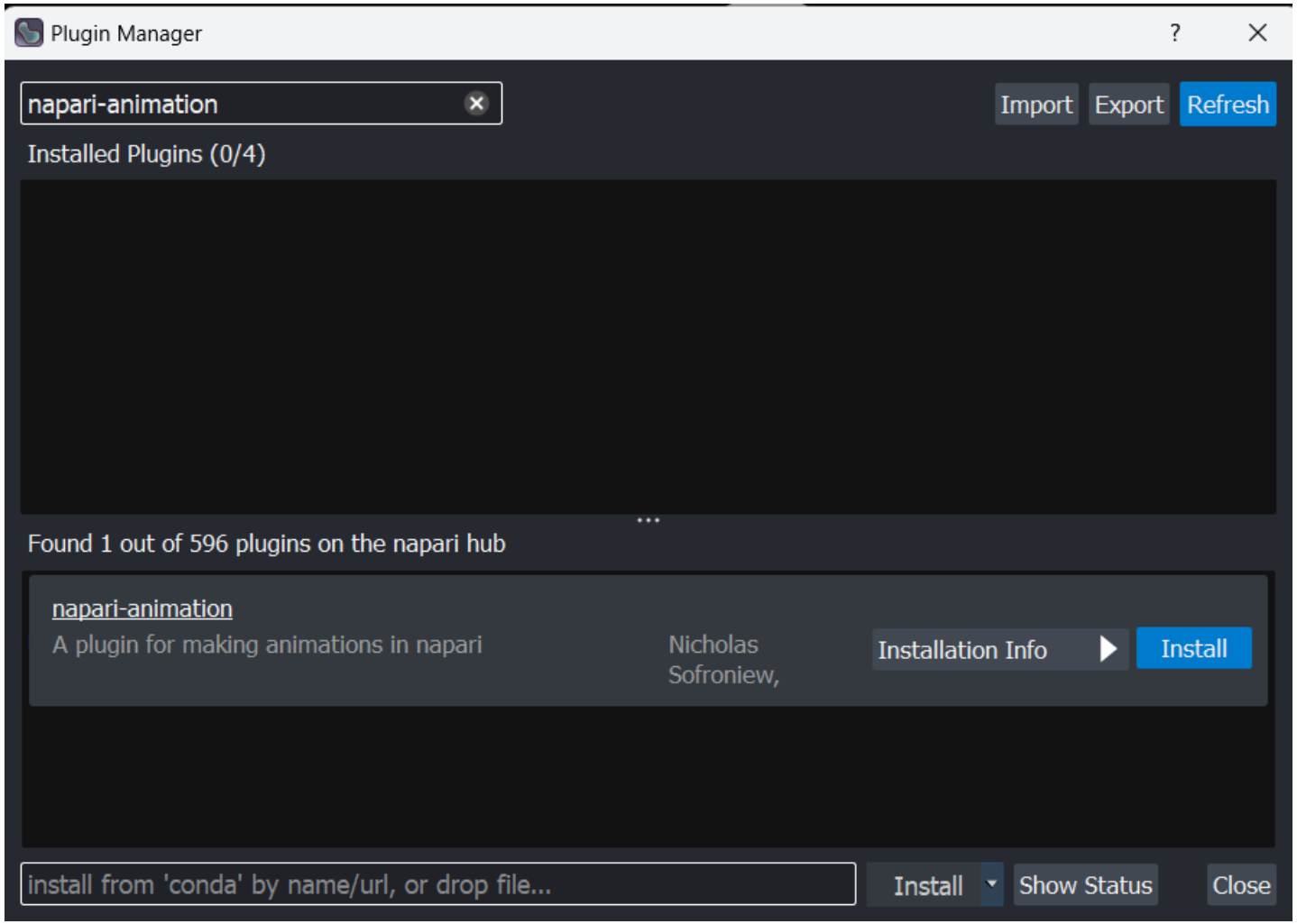
Change n edit dim to set the label in 3D

The editing tools (like the brush) function in 3D, creating a sphere around the cursor, or filling in 3D

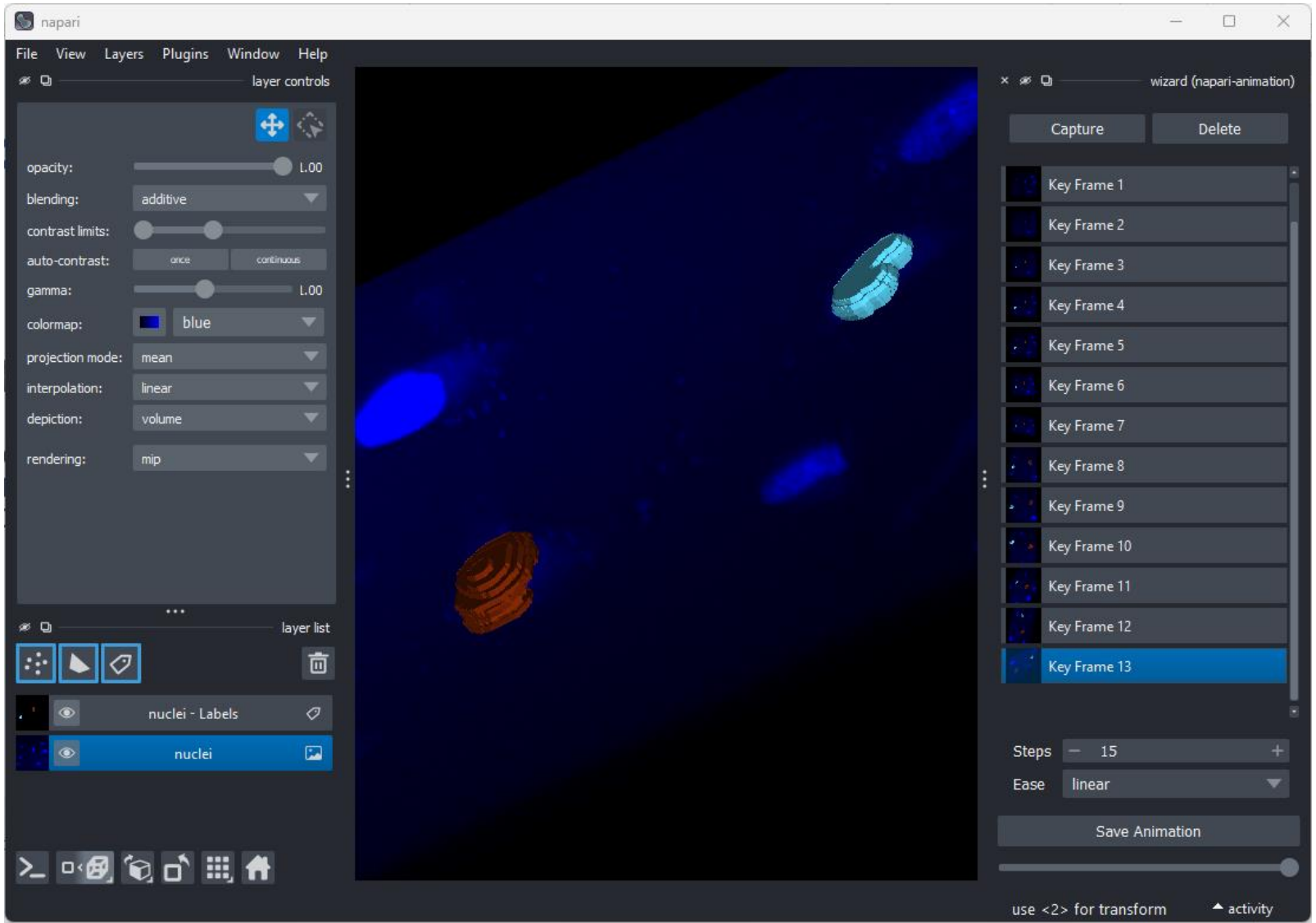


Napari : plugin - napari-animation

https://github.com/napari/napari-animation

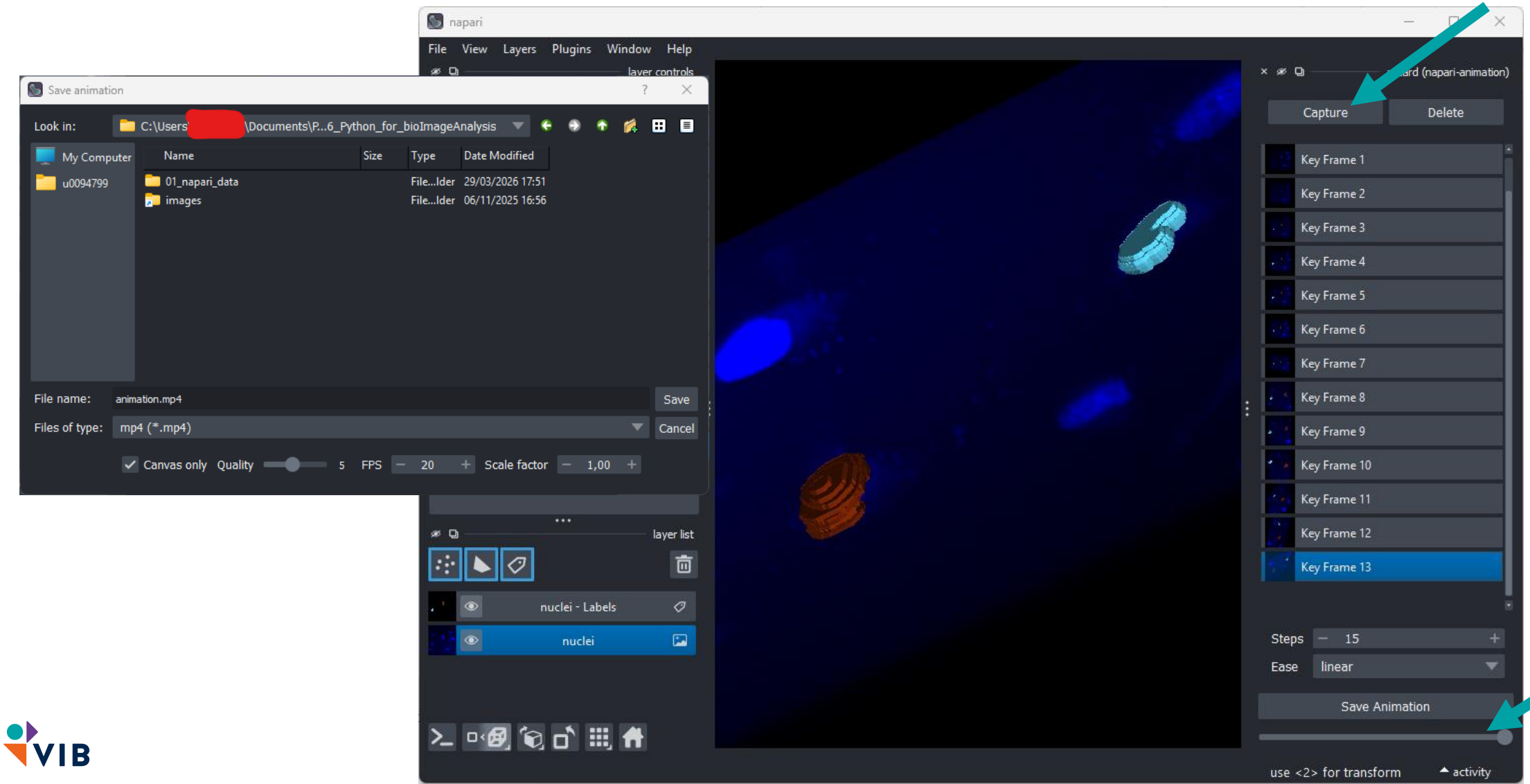


Napari : plugin - napari-animation



Napari : plugin - napari-animation

“Capture” a step as key for the animation (can be label, image. 2D/3D) etc…



Use the slider to preview the animation

Napari : plugin - ndevio

https://github.com/ndev-kit/ndevio

ndevio install a plugin for bioio, a library that allow to open microscopy file format. This library is subdivised in modules. By default, **ndevio** install the base package **bioio**, **bioio-base** and **bioio-ome-tiff**, **bioio-ome-zarr**. You need then to install extra one to open other format, e.g. **bioio-czi** to open Zeiss czi files or **bioio-nd2** to open the Nikon nd2 one.

```
conda activate napari-env
pip install ndevio
pip install bioio-czi
```

Package	Supported File Types
bioio-czi	`CZI`
bioio-dv	`DV`
bioio-imageio	`PNG` , `GIF` , & other similar formats seen here
bioio-lif	`LIF`
bioio-nd2	`ND2`
bioio-ome-tiff	`OME-TIFF` (non-tiled)
bioio-ome-tiled-tiff	`OME TIFF` (tiled)
bioio-ome-zarr	`ZARR`
bioio-sldy	`SLDY`
bioio-tifffile	`TIFF` (non-globbed)
bioio-tiff-glob	`TIFF` (globbed)
bioio-bioformats	Files supported by Bio-Formats (Requires `java` and `maven`, see details)

This plugin support multiple-serie and split automatically the channel, although sometimes the Min/Max intensity is not correct. The scale is also set when exist

VIB

Napari : plugin - napari-bioio-reader

https://github.com/imcf/napari-bioio-reader

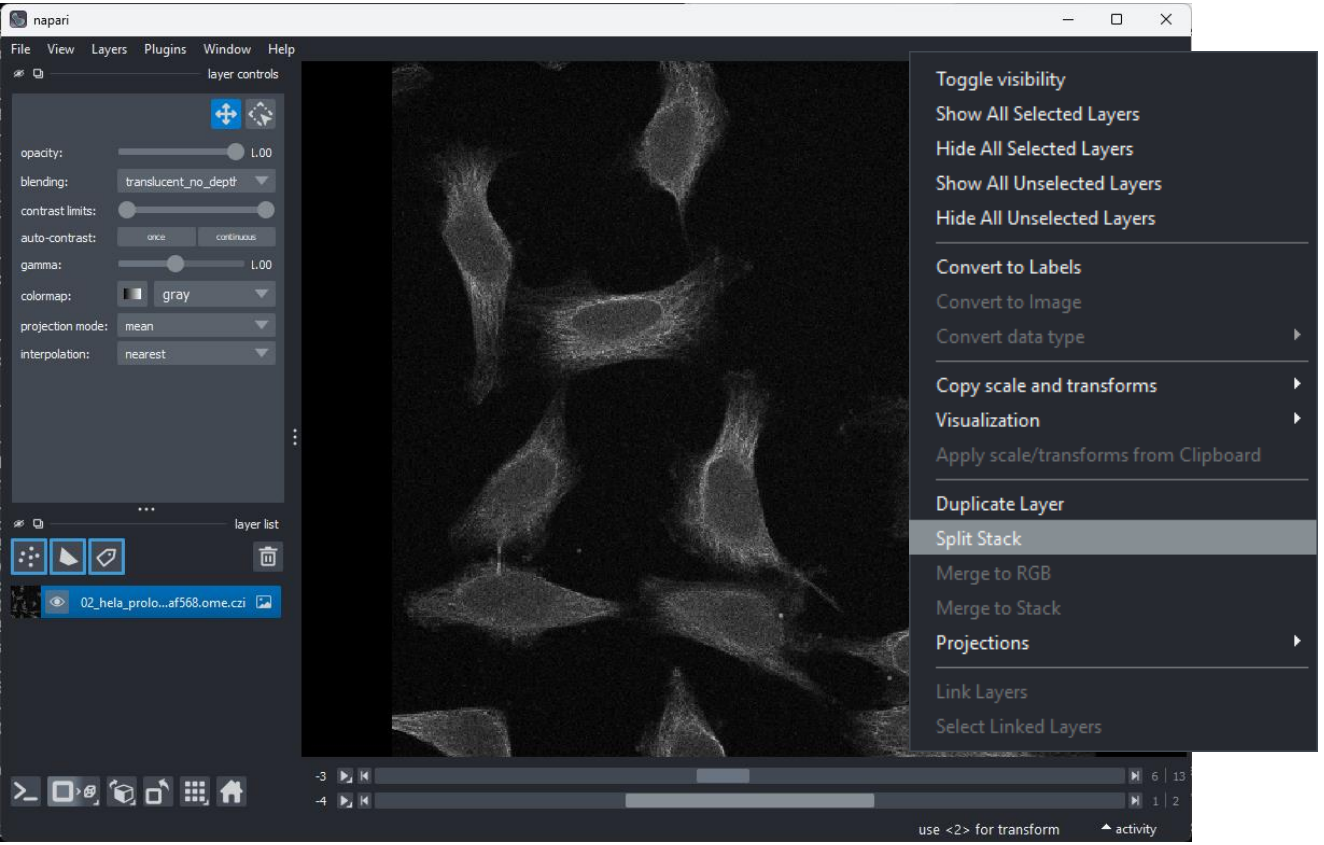
napari-bioio-reader install a plugin for bioio, a library that allow to open microscopy file format. This library is subdivised in modules. By default, **napari-bioio-reader** install the base package **bioio**, **bioio-base** and **bioio-bioformats**. You need then to install extra one to open other format, e.g. **bioio-czi** to open Zeiss czi files or **bioio-nd2** to open the Nikon nd2 one.

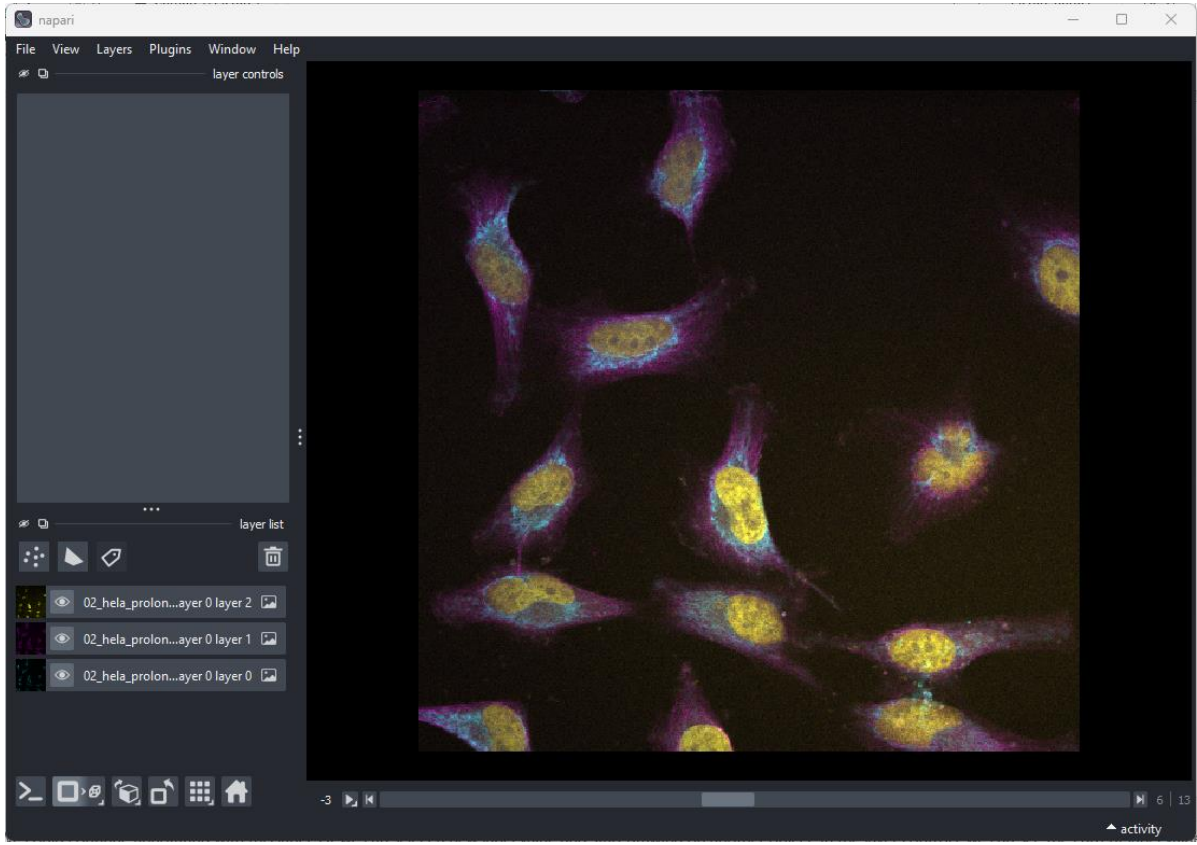
```
conda activate napari-env
pip install napari-bioio-reader
pip install bioio-czi
```

Package	Supported File Types
bioio-czi	`CZI`
bioio-dv	`DV`
bioio-imageio	`PNG` , `GIF` , & other similar formats seen here
bioio-lif	`LIF`
bioio-nd2	`ND2`
bioio-ome-tiff	`OME-TIFF` (non-tiled)
bioio-ome-tiled-tiff	`OME TIFF` (tiled)
bioio-ome-zarr	`ZARR`
bioio-sldy	`SLDY`
bioio-tifffile	`TIFF` (non-globbed)
bioio-tiff-glob	`TIFF` (globbed)
bioio-bioformats	Files supported by Bio-Formats (Requires `java` and `maven`, see details)

This plugin **do no** support multiple-serie and **do not** split automatically the channel, so you need to do it manually and **do not** set the scale

Napari : plugin - napari-bioio-reader

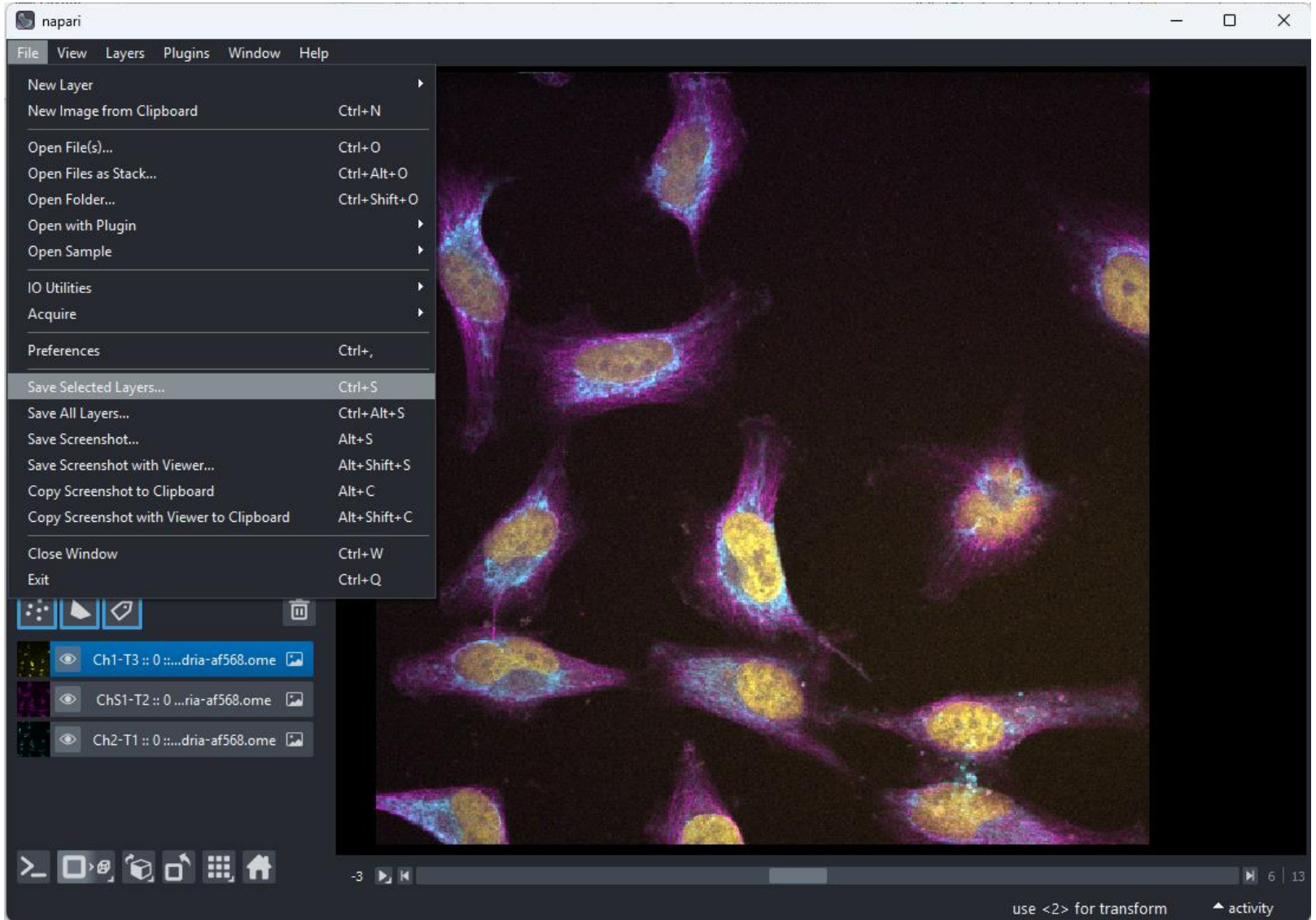




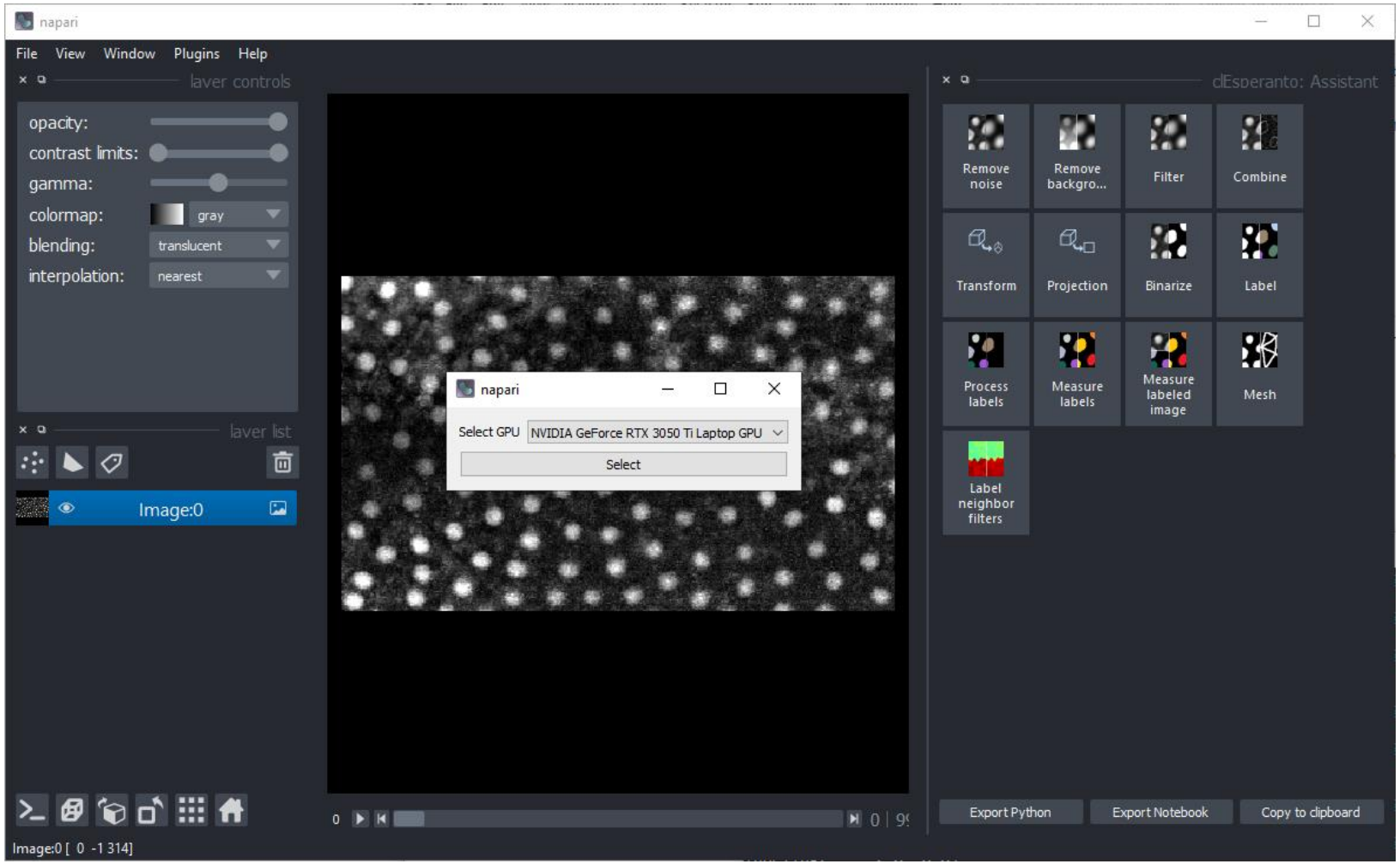
Napari : save layer

`File > Save Selected Layers…`

- Image can be saved as tif
- Label can be saved as tif or png (2D only)
- Points can be saved as CSV

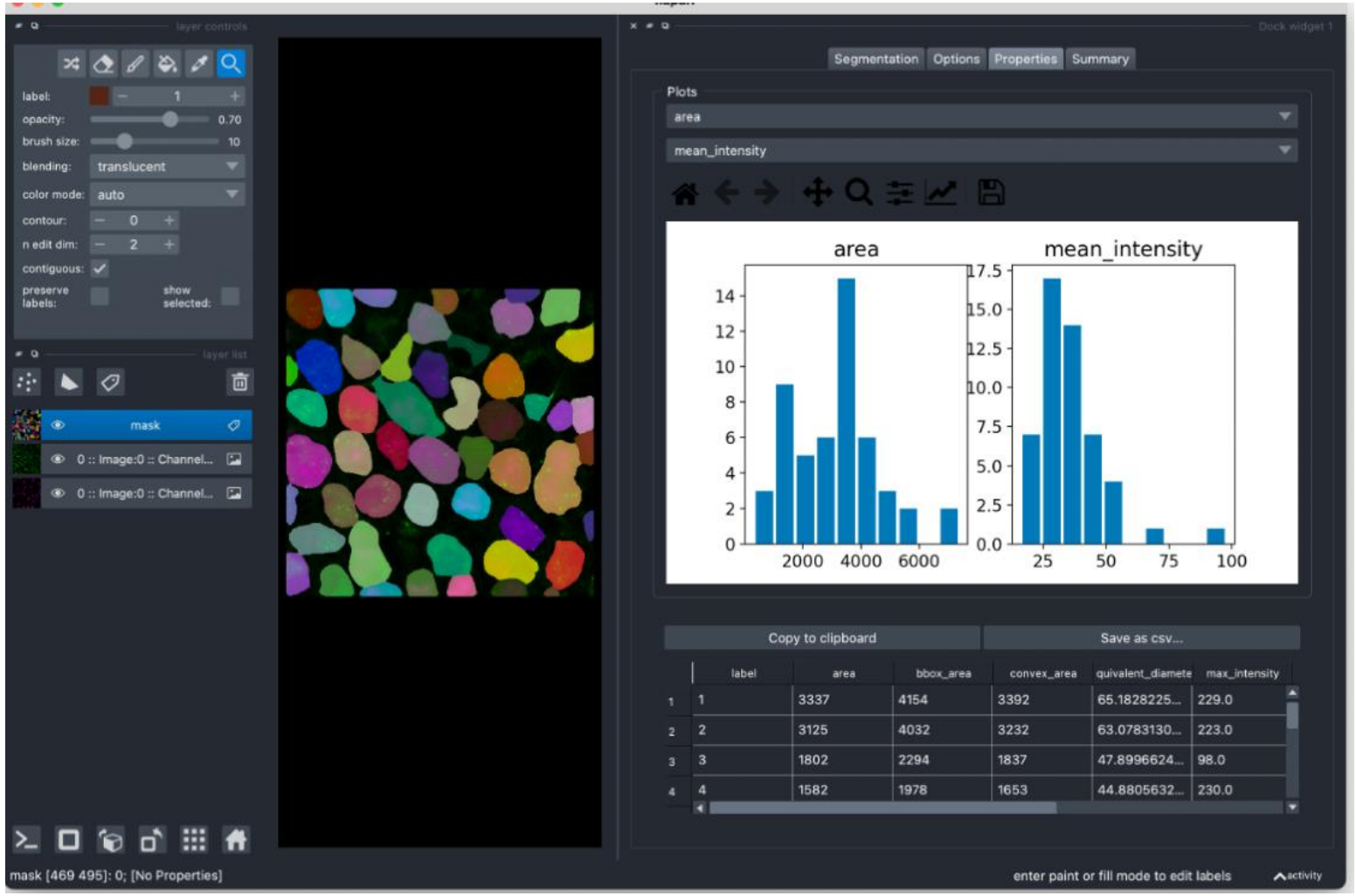


Napari plugin : pyclesperanto assistant



https://github.com/clesperanto/napari_pyclesperanto_assistant

Segmentation : napari-serialcellpose

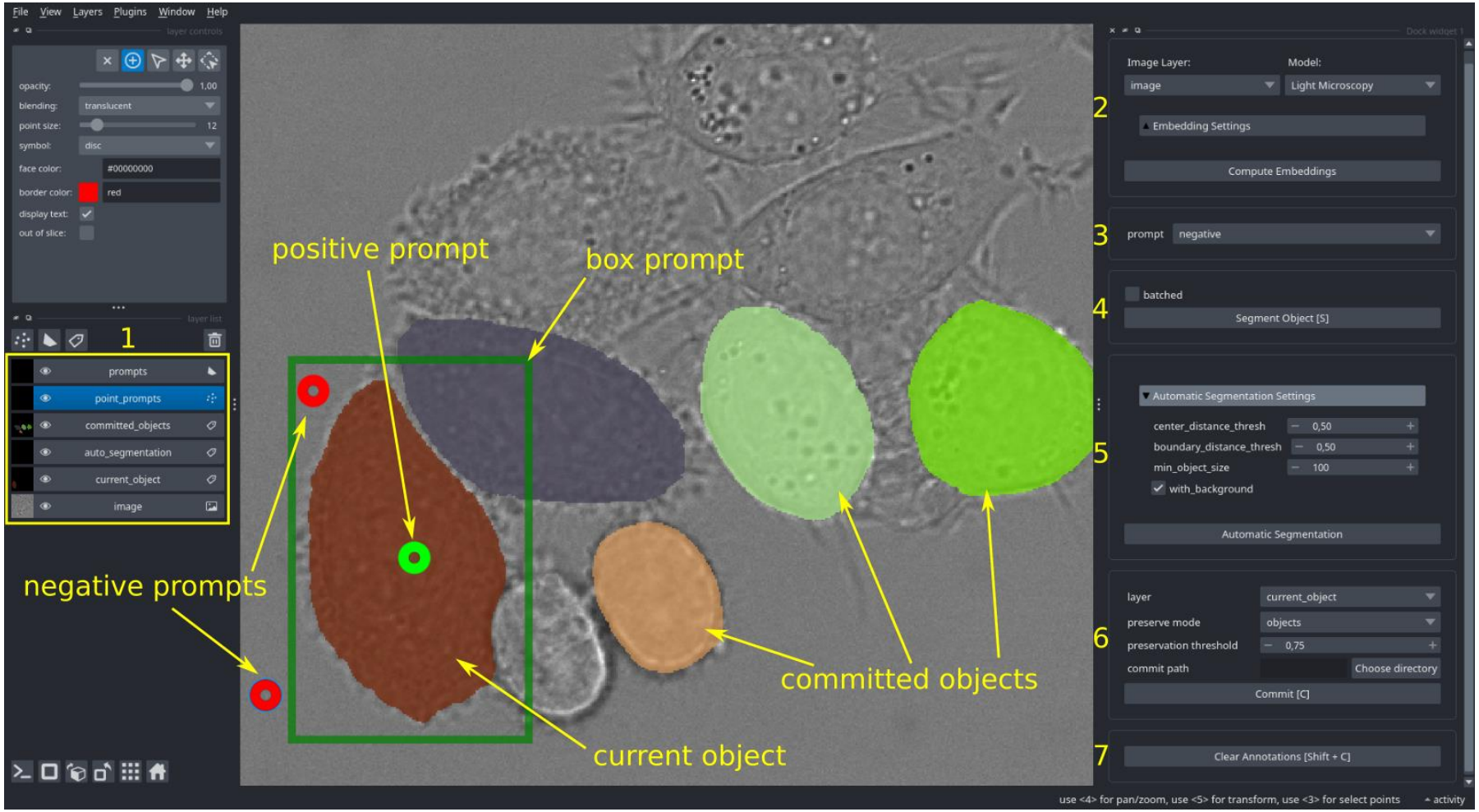


https://github.com/guiwitz/napari-serialcellpose

Segmentation : stardist-napari

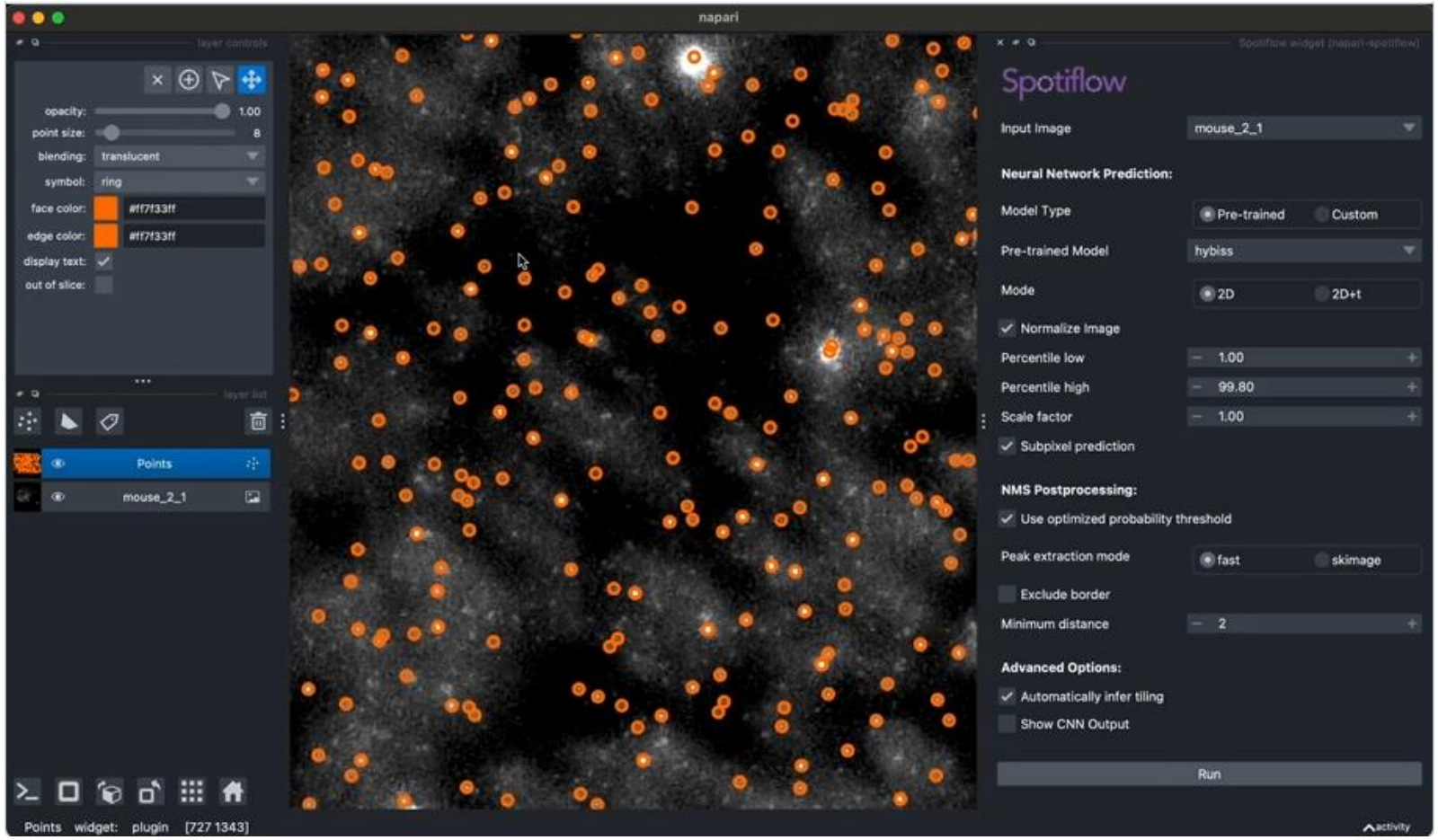
https://github.com/stardist/stardist-napari

Segmentation : micro-sam



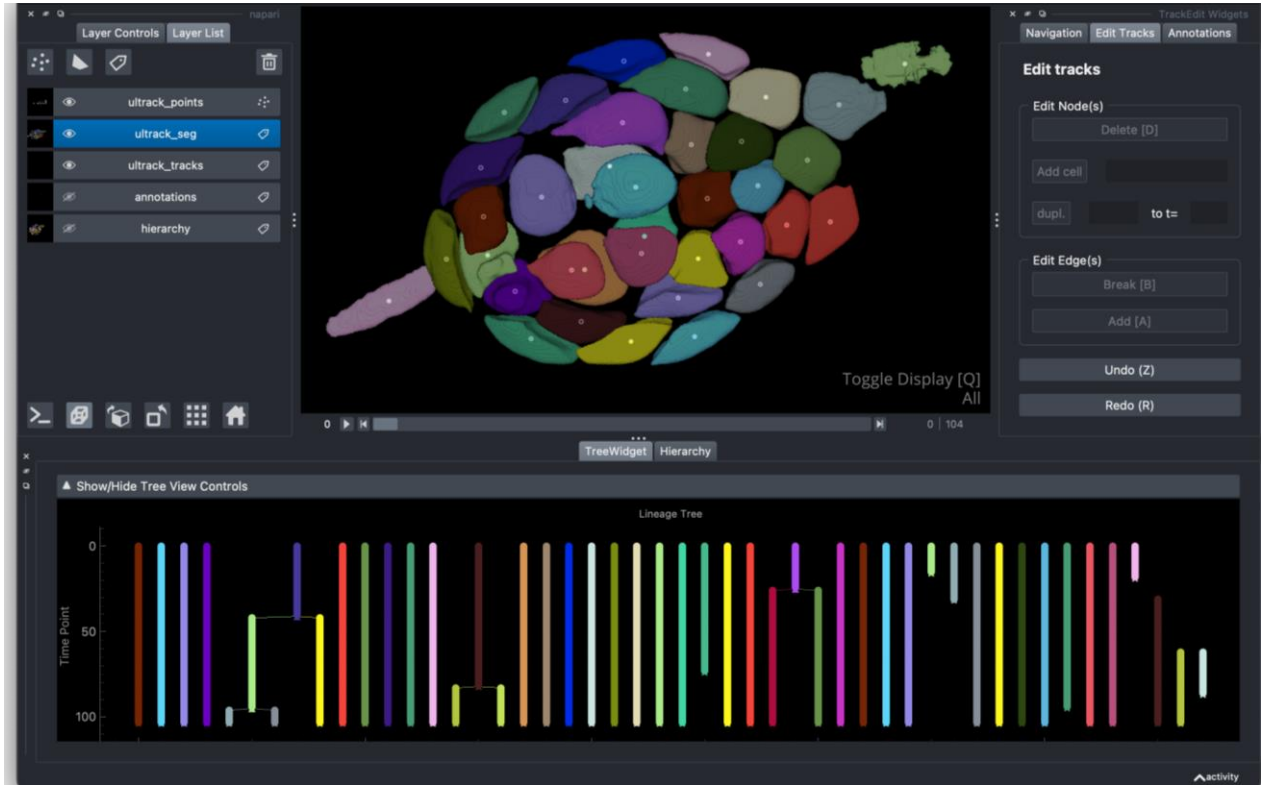
https://github.com/computational-cell-analytics/micro-sam

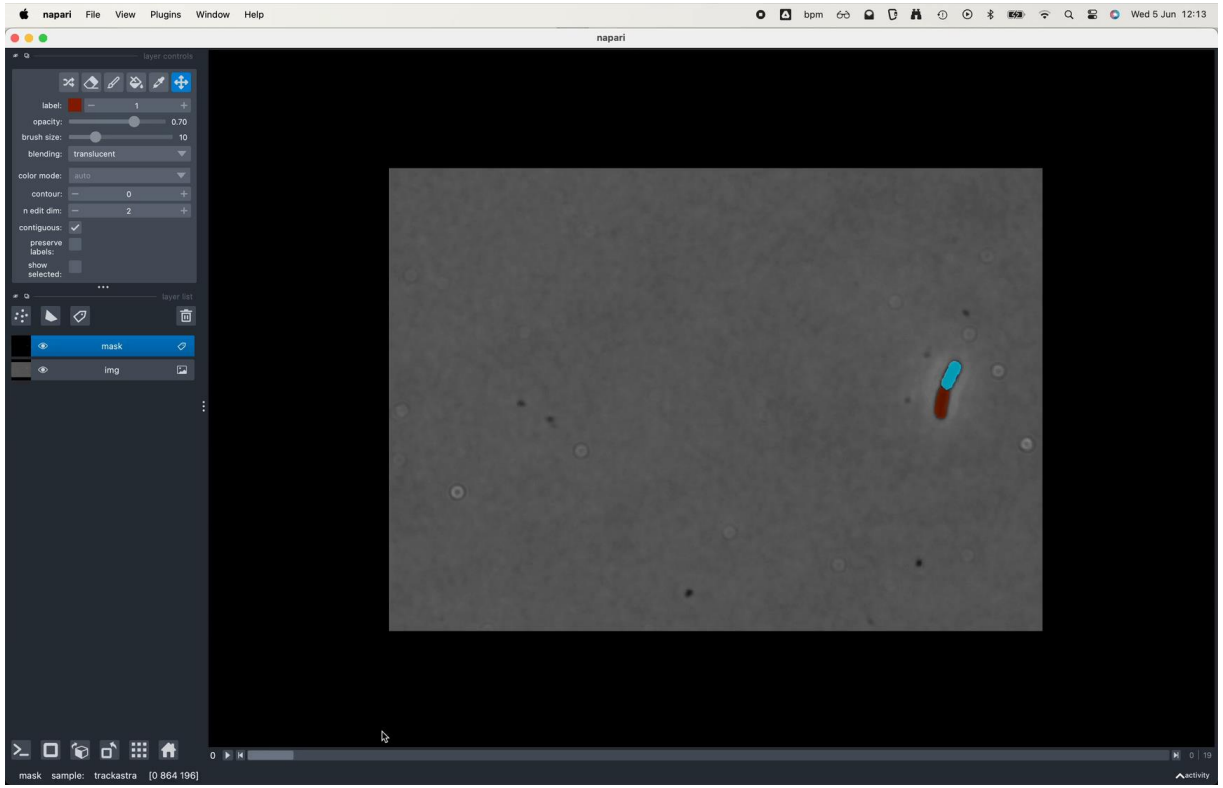
Spot detection : spotiflow



https://github.com/weigertlab/napari-spotiflow

Tracking: ultrack/trackedit and trackastra



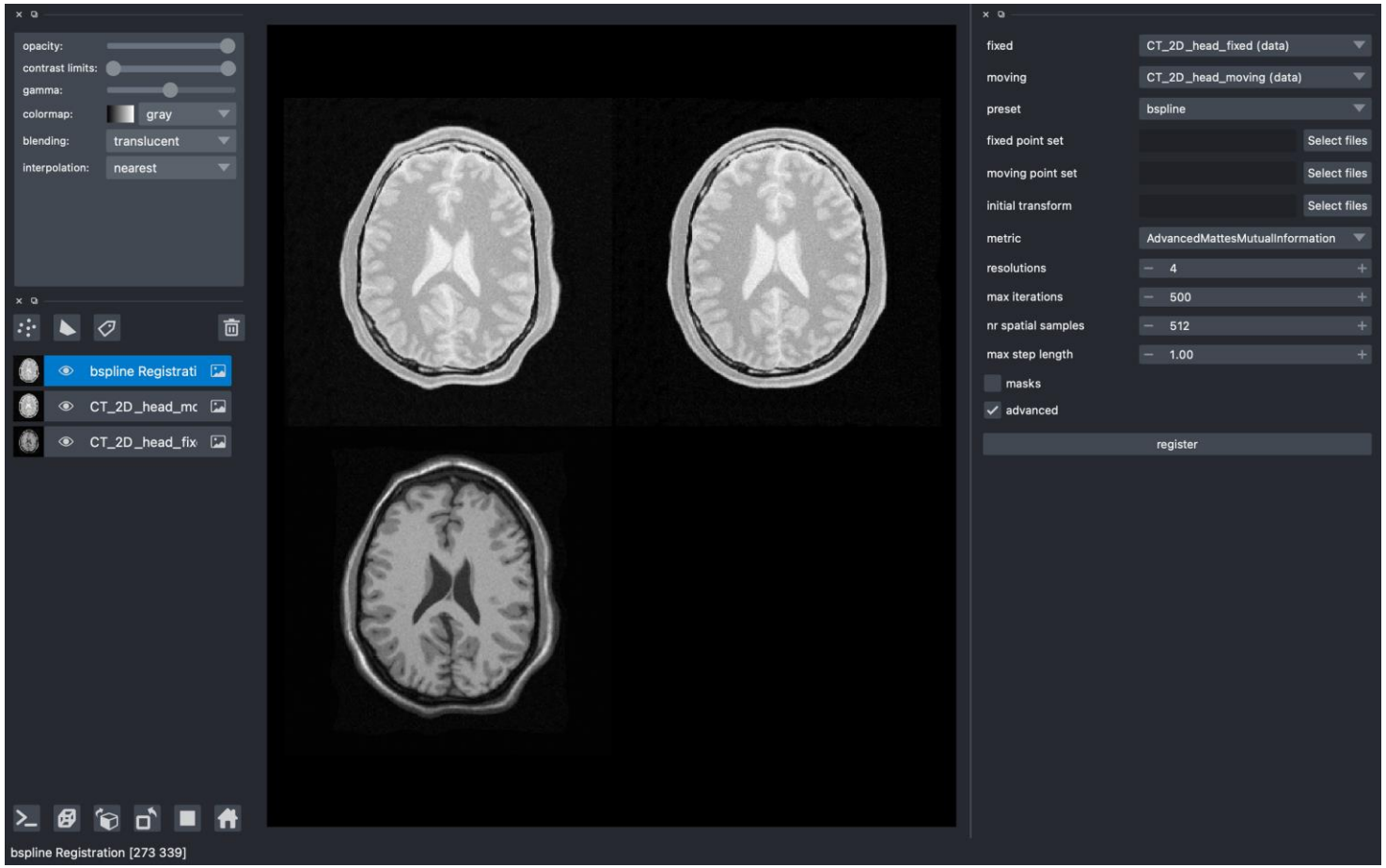


- https://royerlab.github.io/ultrack/napari.html
- https://github.com/royerlab/trackedit

- https://github.com/weigertlab/napari-trackastra

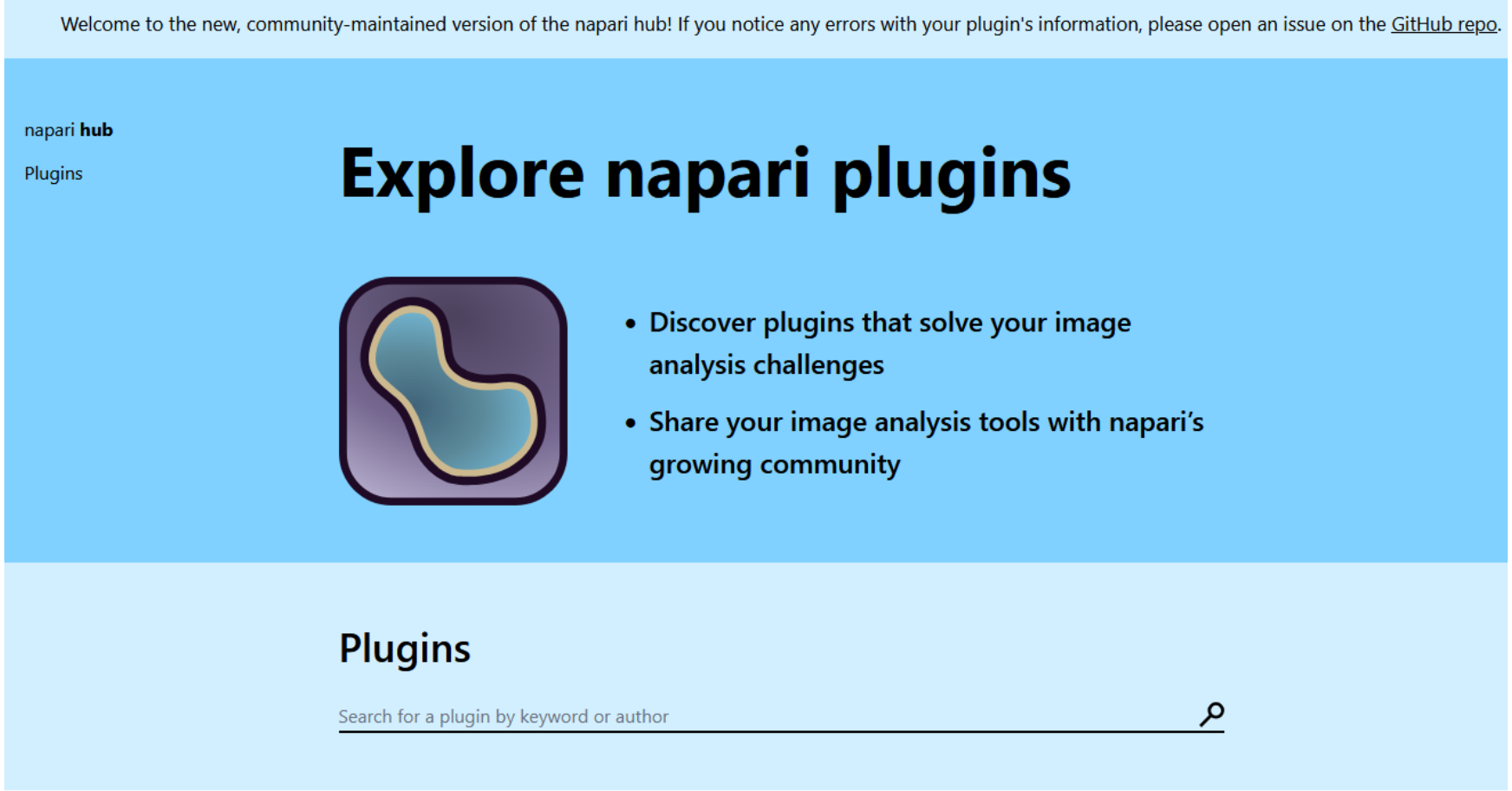
VIB

Registration : elastix-napari



https://github.com/SuperElastix/elastix-napari

And many more… (>600 plugins right now)



Browse plugins: 602

Napari : current issues

Challenge	Core Problem	User Impact
Legacy Engine Removal	Support for `npe1` plugins dropped with version 0.7	Older plugins simply disappear
Library Version Mismatch	**Plugin Incompatibility:** Conflicting requirements	Can't install Tool A and Tool B together (e.g. due to numpy version)
Dependency Conflicts	Core migration to **Pydantic v2**	Incompatible with plugins stuck on v1
Plugin "Abandonware"	Research grant funding expired	Essential tools are no longer updated

Recommendation: Pin to version 0.4.19 for legacy plugin stability. Maybe with an older python version like 3.10

VIB